Undergraduate domestic course

Bachelor of Information Technology (Computer Science)

Year	2027
QUT code	IN01
QTAC code	418801
CRICOS	012656E
Duration	3 years full time 6 years part time
ATAR/Selection rank	70.00
Total credit points	288
Start months	July, February
Domestic fee (indicative)	2027: Fee available from October
Course contact	askqut@qut.edu.au 3138 2000

This PDF contains information about the course structure. For more information about the course see the course information PDF

Course structure

In order to complete this course, you must complete a total of 288 credit points, made up of:

- Core units - 108 credit points
- A Major - 72 credit points
- QUT You - 24 credit points
- Complementary studies - 84 credit points, made up of:
  - a Second Major (72 credit points) and an elective (12 credit points), or
  - a Minor (48 credit points) and electives (36 credit points).

Progression options for your study plan
There are two progression options available, both starting with the same first semester units.
You can choose the pathway that best suits your interests and goals:

Option 1: Common first year
This option allows you to complete a set of foundational core units in your first year before selecting your major. This is the ideal choice if you would like more time to explore your interests before committing to a specific area of study.

Option 2: Major starts in first year
In this option, you begin studying units from your chosen major in your first year. This progression option is suitable if you already know which major you would like to pursue and want to build expertise from the first year.

Regardless of your choice, both options are designed to support a smooth progression through your course.

Structures

- Study plan progression: February & July entry FT - Common first year option
- Study plan progression: February & July entry FT - Major starts in first year option
- Study plan progression: Articulating from IT10 - February & July entry FT
- Study plan progression: Articulating from EE18 or ES21 - February entry FT

Unit Lists

- QUT You
- IT - Major Option Units
- Second Major Options List
- Minor Options List
- University Wide Minor options
- Language Minor options
- Science, Information Technology and Mathematics Undergraduate University Wide Electives

Study plan progression: February & July entry FT - Common first year option

Semesters

- Semester 1 (February) entry
- Year 1, Semester 1
- Year 1, Semester 2
- Year 2, Semester 1
- Year 2, Semester 2
- Year 3, Semester 1
- Year 3, Semester 2
- Semester 2 (July) entry
- Year 1, Semester 2
- Year 2, Semester 1
- Year 2, Semester 2
- Year 3, Semester 1
- Year 3, Semester 2
- Year 4, Semester 1

Code	Title
Semester 1 (February) entry	
Year 1, Semester 1	
IFB102	Introduction to Computer Systems
IFB103	IT Systems Design
IFB104	Introduction to Programming
IFB105	Database Management
Year 1, Semester 2	

Code	Title
IFB201	Introduction to Enterprise Computing
IFB220	Introduction to AI for IT Professionals
IFB240	Cyber Security
Select one QUT You unit (6 credit points)	
Select one QUT You unit (6 credit points)	
Year 2, Semester 1	
CAB201	Object-Oriented Programming and Design
CS Elective or Complementary studies unit	
Complementary studies unit	
Complementary studies unit	
Year 2, Semester 2	
CAB222	Networks
CAB302	Agile Software Engineering
CS Elective or Complementary studies unit	
Complementary studies unit	
Year 3, Semester 1	
IFB398	IT Capstone Project (Phase 1)
IFB320	Generative AI
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Year 3, Semester 2	
IFB399	IT Capstone Project (Phase 2)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	
Semester 2 (July) entry	
Year 1, Semester 2	
IFB102	Introduction to Computer Systems
IFB103	IT Systems Design
IFB104	Introduction to Programming
IFB105	Database Management
Year 2, Semester 1	
IFB201	Introduction to Enterprise Computing
IFB220	Introduction to AI for IT Professionals
IFB240	Cyber Security
Select one QUT You unit (6 credit points)	
Select one QUT You unit (6 credit points)	
Year 2, Semester 2	
CAB201	Object-Oriented Programming and Design
CAB222	Networks
CS Elective unit or Complementary studies unit	
Complementary studies unit	
Year 3, Semester 1	
IFB320	Generative AI
CAB302	Agile Software Engineering
CS Elective unit or Complementary studies unit	
Complementary studies unit	
Year 3, Semester 2	
IFB398	IT Capstone Project (Phase 1)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	

Code	Title
Complementary studies unit	
Year 4, Semester 1	
IFB399	IT Capstone Project (Phase 2)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	

Study plan progression: February & July entry FT - Major starts in first year option

Semesters

- Semester 1 (February) entry
- Year 1, Semester 1
- Year 1, Semester 2
- Year 2, Semester 1
- Year 2, Semester 2
- Year 3, Semester 1
- Year 3, Semester 2
- Semester 2 (July) entry
- Year 1, Semester 2
- Year 2, Semester 1
- Year 2, Semester 2
- Year 3, Semester 1
- Year 3, Semester 2
- Year 4, Semester 1

Code	Title
Semester 1 (February) entry	
Year 1, Semester 1	
IFB102	Introduction to Computer Systems
IFB103	IT Systems Design
IFB104	Introduction to Programming
IFB105	Database Management
Year 1, Semester 2	
IFB220	Introduction to AI for IT Professionals
IFB240	Cyber Security
CAB201	Object-Oriented Programming and Design
Select one QUT You unit (6 credit points)	
Select one QUT You unit (6 credit points)	
Year 2, Semester 1	
IFB201	Introduction to Enterprise Computing
IFB320	Generative AI
CS Elective or Complementary studies unit	
Complementary studies unit	
Year 2, Semester 2	
CAB222	Networks
CAB302	Agile Software Engineering
CS Elective or Complementary studies unit	
Complementary studies unit	
Year 3, Semester 1	
IFB398	IT Capstone Project (Phase 1)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	
Year 3, Semester 2	
IFB399	IT Capstone Project (Phase 2)

Code	Title
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	
Semester 2 (July) entry	
Year 1, Semester 2	
IFB102	Introduction to Computer Systems
IFB103	IT Systems Design
IFB104	Introduction to Programming
IFB105	Database Management
Year 2, Semester 1	
IFB220	Introduction to AI for IT Professionals
IFB240	Cyber Security
CAB201	Object-Oriented Programming and Design
Select one QUT You unit (6 credit points)	
Select one QUT You unit (6 credit points)	
Year 2, Semester 2	
IFB201	Introduction to Enterprise Computing
CAB222	Networks
CS Elective unit or Complementary studies unit	
Complementary studies unit	
Year 3, Semester 1	
IFB320	Generative AI
CAB302	Agile Software Engineering
CS Elective unit or Complementary studies unit	
Complementary studies unit	
Year 3, Semester 2	
IFB398	IT Capstone Project (Phase 1)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	
Year 4, Semester 1	
IFB399	IT Capstone Project (Phase 2)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	

Study plan progression: Articulating from IT10 - February & July entry FT

Semesters

- IT10 advance standing (96 credit points)
- Semester 1 (February) entry
- Year 1, Semester 1
- Year 1, Semester 2
- Year 2, Semester 1
- Year 2, Semester 2
- Semester 2 (July) entry
- Year 1, Semester 2
- Year 2, Semester 1
- Year 2, Semester 2
- Year 3, Semester 1

Code	Title
IT10 advance standing (96 credit points)	
72 credit points of core units (IFB102, IFB103, IFB104, IFB105, IFB201, IFB240)	
12 credit points of complementary studies units	
12 credit points of QUT You units	
Semester 1 (February) entry	
Year 1, Semester 1	
IFB220	Introduction to AI for IT Professionals
CAB201	Object-Oriented Programming and Design
CS Elective or Complementary studies unit	
Complementary studies unit	
Year 1, Semester 2	
CAB222	Networks
CAB302	Agile Software Engineering
CS Elective or Complementary studies unit	
Complementary studies unit	
Year 2, Semester 1	
IFB398	IT Capstone Project (Phase 1)
IFB320	Generative AI
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Year 2, Semester 2	
IFB399	IT Capstone Project (Phase 2)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	
Semester 2 (July) entry	
Year 1, Semester 2	
CAB201	Object-Oriented Programming and Design
CAB222	Networks
IFB220	Introduction to AI for IT Professionals
CS Elective unit or Complementary studies unit	
Year 2, Semester 1	
IFB320	Generative AI
CAB302	Agile Software Engineering
CS Elective unit or Complementary studies unit	
Complementary studies unit	
Year 2, Semester 2	
IFB398	IT Capstone Project (Phase 1)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	
Year 3, Semester 1	
IFB399	IT Capstone Project (Phase 2)
CS Elective unit or 2 x 6cps of QUT You units	
Complementary studies unit	
Complementary studies unit	

Study plan progression: Articulating from EE18 or ES21 - February entry FT

Semesters

- EE18 or ES21 advance standing (96 credit points)
- Semester 1 (February) entry
- Year 1, Semester 1
- Year 1, Semester 2

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- Year 2, Semester 1
- Year 2, Semester 2

Code	Title
EE18 or ES21 advance standing (96 credit points)	
48 credit points of core units (IFB102, IFB103, IFB104, IFB105)	
36 credit points of complementary studies units (for the completion of EED001, EED002, EED003, EED004 in EE18 OR ESD001, ESD002, ESD003, ESD004 in ES21)	
12 credit points of QUT You units	
Semester 1 (February) entry	
Year 1, Semester 1	
IFB220	Introduction to AI for IT Professionals
IFB240	Cyber Security
CAB201	Object-Oriented Programming and Design
Complementary studies unit	
Year 1, Semester 2	
CAB302	Agile Software Engineering
CAB222	Networks
CS Elective unit	
Complementary studies unit	
Year 2, Semester 1	
IFB398	IT Capstone Project (Phase 1)
IFB320	Generative AI
CS Elective unit	
Complementary studies unit	
Year 2, Semester 2	
IFB399	IT Capstone Project (Phase 2)
IFB201	Introduction to Enterprise Computing
2 x 6cps of QUT You units	
Complementary studies unit	

QUT You

QUT You	
Code	Title
QUT001	QUT You: Artificial Intelligence in the Real World
QUT002	QUT You: Walking on Country
QUT003	QUT You: Real Action for Real Change
QUT004	QUT You: Living and Working Collaboratively, Ethically, and Inclusively
QUT005	QUT You: Seeing Me, Seeing You: Skills for a Diverse World
QUT006	QUT You: The Art of Pitching
QUT007	QUT You: Fighting 'Fake News'
QUT008	QUT You: Think Like a Computer and Change the World
QUT009	QUT You: Data Science for Society
QUT010	QUT You: People with Robots
QUT013	QUT You: Human Agency in the Digital World

IT - Major Option Units

From 2026 onwards, the Bachelor of Information Technology majors—Computer Science, Enterprise Computing, and Process Analytics and Automation — include optional units within each major.

In this list

- Computer Science Major
- Enterprise Computing Major
- Process Analytics and Automation Major

Computer Science Major	
Code	Title
IN01MJR-COMSCN v1	
Computer Science Major General Option Units List	
CAB210	User Experience Fundamentals
CAB230	Web Computing
CAB301	Algorithms and Complexity
CAB330	Machine Learning for Decision Making
CAB401	High Performance and Parallel Computing
CAB403	Systems Programming
CAB420	Machine Learning
CAB432	Cloud Computing
IFB343	Secure Software Development
IFB452	Blockchain Technology
Computer Science Major Advanced Option Units List	
CAB401	High Performance and Parallel Computing
CAB403	Systems Programming
CAB420	Machine Learning
CAB432	Cloud Computing
IFB452	Blockchain Technology

Enterprise Computing Major	
Code	Title
IN01MJR-ENTCOM v1	
Enterprise Computing Major Option Units	
IAB201	Modelling Techniques for Information Systems
IAB204	Business Analysis for IT Systems
IAB260	Social Technologies
IAB305	IT Strategy and Management
IAB321	Process Technologies
IAB353	Data Analytics for Enterprise Systems
IAB402	IT Consulting and Leadership
IFB320	Generative AI
IFB452	Blockchain Technology

Process Analytics and Automation Major	
Code	Title
IN01MJR-PANAUT v1	
Process Analytics and Automation Major Option Units	
DSB201	Advanced Databases
IAB204	Business Analysis for IT Systems
IAB353	Data Analytics for Enterprise Systems

Second Major Options List

Below is a list of available Second Major options for selection in 2026 onwards. Please ensure that your selection of Second Major is different from your Primary Major.

* Any unit(s) that is core in your primary major, can NOT be selected in your chosen second major.

In this list

Artificial Intelligence Second Major	
Code	Title
IN01SMJ-ARTINT v1	
Core Units	
CAB330	Machine Learning for Decision Making
CAB420	Machine Learning
DSB102	Introduction to Machine Learning
CAB201 or a unit from the Information Technology options list. You must select CAB201 if it's not in your first major.	
CAB201	Object-Oriented Programming and Design
Information Technology unit options list	
IAB353 or a unit from the Information Technology unit options list. You must select IAB353 if it's not in your first major.	
IAB353	Data Analytics for Enterprise Systems
Information Technology unit options list	
IFB320 or a unit from the Information Technology unit options list. You must select IFB320 if it's not in your first major.	
IFB320	Generative AI
Information Technology unit options list	

Business Analysis and IT Management Second Major	
Code	Title
IN01SMJ-BAITMG v1	
Core Units	
IAB204	Business Analysis for IT Systems
IAB305	IT Strategy and Management
IAB402	IT Consulting and Leadership
IAB203 or a unit from the Information Technology options list. You must select IAB203 if it's not in your first major.	
IAB203	Process Modelling
Information Technology unit options list	
IAB353 or a unit from the Information Technology options list. You must select IAB353 if it's not in your first major.	
IAB353	Data Analytics for Enterprise Systems
Information Technology unit options list	
IAB401 or a unit from the Information Technology options list. You must select IAB401 if it's not in your first major.	
IAB401	Enterprise Architecture
Information Technology unit options list	

Computer Science Second Major	
Code	Title
IN01SMJ-COMSCN v1	
CAB201 or a unit from the Information Technology options list. You must select CAB201 if it's not in your first major.	
CAB201	Object-Oriented Programming and Design
Information Technology unit options list	
CAB222 or a unit from the Information Technology options list. You must select CAB222 if it's not in your first major.	
CAB222	Networks
Information Technology unit options list	
CAB302 or a unit from the Information Technology options list. You must select CAB302 if it's not in your first major.	
CAB302	Agile Software Engineering
Information Technology unit options list	
IFB320 or a unit from the Information Technology options list. You must select IFB320 if it's not in your first major.	
IFB320	Generative AI
Information Technology unit options list	
Computer Science Second Major General Option Units List. Select one unit from the Computer Science Second Major General Option Units List.	
CAB210	User Experience Fundamentals
CAB230	Web Computing
CAB301	Algorithms and Complexity
CAB330	Machine Learning for Decision Making
CAB401	High Performance and Parallel Computing
CAB403	Systems Programming
CAB420	Machine Learning
CAB432	Cloud Computing
IFB343	Secure Software Development
IFB452	Blockchain Technology
Computer Science Second Major Advanced Option Units List. Select one unit from the Computer Science Second Major General Option Units List	
CAB401	High Performance and Parallel Computing
CAB403	Systems Programming
CAB420	Machine Learning
CAB432	Cloud Computing
IFB452	Blockchain Technology

Cyber Security Second Major	
Code	Title
IN01SMJ-CYBSEC v1	
Core units	
CAB340	Cryptography
CAB443	Systems Security
CAB444	Secure Network Architectures
IAB246	Organisations and Security: Governance, Risk and Compliance
IFB343	Secure Software Development
CAB222 or a unit from the Information Technology options list. You must select CAB222 if it's not in your first major.	
CAB222	Networks
Information Technology unit options list	

Enterprise Computing Second Major	
Code	Title
IN01SMJ-ENTCOM v1	
Core units	

Enterprise Computing Second Major	
IAB207	Rapid Web Application Development
IAB330	Applied Internet of Things
CAB201 or a unit from the Information Technology options list. You must select CAB201 if it's not in your first major.	
CAB201	Object-Oriented Programming and Design
Information Technology unit options list	
IAB251 or a unit from the Information Technology options list. You must select IAB251 if it's not in your first major.	
IAB251	Software Engineering for Enterprise Systems
Information Technology unit options list	
IAB401 or a unit from the Information Technology options list. You must select IAB401 if it's not in your first major.	
IAB401	Enterprise Architecture
Information Technology unit options list	
Enterprise Computing Option units. Select one unit from the Enterprise Computing Option units list	
IAB201	Modelling Techniques for Information Systems
IAB204	Business Analysis for IT Systems
IAB260	Social Technologies
IAB305	IT Strategy and Management
IAB321	Process Technologies
IAB353	Data Analytics for Enterprise Systems
IAB402	IT Consulting and Leadership
IFB320	Generative AI
IFB452	Blockchain Technology

Process Analytics and Automation Second Major	
Code	Title
IN01SMJ-PANAUT v1	
Core units	
DSB202	Data Ethics and Society
IAB201	Modelling Techniques for Information Systems
IAB320	Process Improvement
IAB321	Process Technologies
IAB203 or a unit from the Information Technology options list. You must select IAB203 if it's not in your first major.	
IAB203	Process Modelling
Information Technology unit options list	
Process Analytics and Automation option units	
DSB201	Advanced Databases
IAB402	IT Consulting and Leadership
IAB353	Data Analytics for Enterprise Systems

Software Development Second Major	
Code	Title
IN01SMJ-SOFDEV v1	
Core units	
CAB230	Web Computing
CAB301	Algorithms and Complexity
CAB432	Cloud Computing
CAB201 or a unit from the Information Technology options list. You must select CAB201 if it's not in your first major.	
CAB201	Object-Oriented Programming and Design
Information Technology unit options list	
CAB302 or 12cp from the Information Technology unit options list. You must select CAB302 if it's not in your first major.	
CAB302	Agile Software Engineering
Information Technology unit options list	
IAB251 or a unit from the Information Technology options list. You must select IAB251 if it's not in your first major.	
IAB251	Software Engineering for Enterprise Systems
Information Technology unit options list	

Minor Options List

Below is a list of Minors available for selection in 2026 onwards. Please ensure that your selection of Minor is different from your Primary Major.

- Please refer to the study sequences to plan your program. You must complete 48 credit points (normally four 12 credit point units) from the specified units to achieve a minor, following semester of offer and unit prerequisites (where applicable) to determine the order of enrolment.
- Any unit(s) that is mandatory in your primary major, can NOT be selected in your chosen minors.
- Any unit(s) that appear in multiple minors can only contribute towards the completion of one of these minors.

In this list

Artificial Intelligence Minor	
Code	Title
IN01MNR-ARTINT v1	
Complete 48 credit points of study (four units) from the Artificial Intelligence Minor Option Units list	
CAB201	Object-Oriented Programming and Design
CAB330	Machine Learning for Decision Making
CAB420	Machine Learning
DSB102	Introduction to Machine Learning
IAB353	Data Analytics for Enterprise Systems
IFB320	Generative AI

Business Analysis and IT Management Minor	
Code	Title
IN01MNR-BAITMG v1	
Core unit	
IAB204	Business Analysis for IT Systems
IAB203 or a unit from the Information Technology options list. You must select IAB203 if it's not in your first major.	
IAB203	Process Modelling
Information Technology unit options list	
IAB353 or a unit from the Information Technology options list.	

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Business Analysis and IT Management Minor

You must select IAB353 if it's not in your first major.	
IAB353	Data Analytics for Enterprise Systems
Information Technology unit options list.	
Business Analysis and IT Management Minor Option Units. Select one unit from the Business Analysis and IT Management Minor Option Units list.	
IAB305	IT Strategy and Management
IAB401	Enterprise Architecture
IAB402	IT Consulting and Leadership

Computational Mathematics and Simulation Minor

Code	Title
IN01MNR-COMASI v1	
Core units	
MXB103	Introductory Computational Mathematics
MXB161	Computational Explorations
MXB261	Game Theory and Simulation for Decision-Making
Computational Mathematics and Simulation Minor Option Units. Complete 12 credit points of study (any one unit) from the Computational Mathematics and Simulation Option Units list. Students who have not completed high school Specialist Mathematics or equivalent should choose MXB100.	
MXB100	Introductory Calculus and Algebra
MXB105	Calculus and Differential Equations

Computer Science Minor

Code	Title
IN01MNR-COMPSC	
Complete 48 credit points of study (four units) from the Computer Science Minor Option Units List	
CAB201	Object-Oriented Programming and Design
CAB210	User Experience Fundamentals
CAB230	Web Computing
CAB222	Networks
CAB301	Algorithms and Complexity
CAB302	Agile Software Engineering
CAB330	Machine Learning for Decision Making
CAB401	High Performance and Parallel Computing
CAB403	Systems Programming
CAB420	Machine Learning
CAB432	Cloud Computing
IFB320	Generative AI
IFB343	Secure Software Development
IFB452	Blockchain Technology
Note: CAB202, CAB203 are permitted to count towards this Minor if completed in 2025 or prior.	
Note: CAB303 is permitted to count towards this Minor if completed in 2022 or prior.	

Cyber Security Minor

Code	Title
IN01MNR-CYBSEC v1	
Core units	
IAB246	Organisations and Security: Governance, Risk and Compliance
CAB443	Systems Security
CAB222 or a unit from the Information Technology options list. You must select CAB222 if it's not in your first major.	
CAB222	Networks
Information Technology unit options list	
Cyber Security Minor Option Units. Select one unit from the Cyber Security Minor Option Units list	
CAB340	Cryptography
CAB444	Secure Network Architectures
IFB343	Secure Software Development

Data Science Minor

Code	Title
IN01MNR-DATSCI v1	
Core unit	
DSB100	Fundamentals of Data Science
Data Science Minor Option units. Complete 36 credit points of study (any three units) from the Data Science Option Units list	
MXB107	Probability and Statistics
DSB200	Applied Data Science
DSB201	Advanced Databases
DSB202	Data Ethics and Society
CAB330	Machine Learning for Decision Making
IFB320	Generative AI

Enterprise Computing Minor

Code	Title
IN01MNR-ENTCOM v1	
CAB201 or a unit from the Information Technology options list. You must select CAB201 if it's not in your first major.	
CAB201	Object-Oriented Programming and Design
Information Technology unit options list	
IAB251 or a unit from the Information Technology options list. You must select IAB251 if it's not in your first major.	
IAB251	Software Engineering for Enterprise Systems
Information Technology unit options list	
IAB401 or a unit from the Information Technology options list. You must select IAB401 if it's not in your first major.	
IAB401	Enterprise Architecture
Information Technology unit options list.	
Enterprise Computing Minor Option Units. Select one unit from the Enterprise Computing Minor Option Units list	
IAB201	Modelling Techniques for Information Systems
IAB204	Business Analysis for IT Systems
IAB207	Rapid Web Application Development
IAB260	Social Technologies
IAB305	IT Strategy and Management
IAB321	Process Technologies
IAB330	Applied Internet of Things
IAB353	Data Analytics for Enterprise Systems
IAB402	IT Consulting and Leadership
IFB320	Generative AI

Enterprise Computing Minor	
IFB452	Blockchain Technology

Human-Centred Computing Minor	
Code	Title
IN01MNR-HUMCOM v1	
Core units	
CAB210	User Experience Fundamentals
CAB310	Interaction and Experience Design
Human-Centred Computing Minor Option Units. Complete 24 credit points of study (any two units) from the Human-Centred Computing Core Option Units list.	
CAB230	Web Computing
IAB230	Design of Enterprise IoT
IAB260	Social Technologies
IGB120	Introduction to Game Design
DXB212	Tangible Interaction Design

Internet of Things Minor	
Code	Title
IN01MNR-INTHGS v1	
Core Units	
IAB230	Design of Enterprise IoT
IAB330	Applied Internet of Things
Internet of Things Minor Option Units. Complete 24 credit points of study (any two units) from the Internet of Things Minor Option Units list	
CAB210	User Experience Fundamentals
CAB230	Web Computing
CAB310	Interaction and Experience Design
CAB432	Cloud Computing
IAB402	IT Consulting and Leadership
IFB452	Blockchain Technology

Process Analytics and Automation Minor	
Code	Title
IN01MNR-PANAUT v1	
Core Units	
IAB320	Process Improvement
IAB321	Process Technologies
IAB203 or a unit from the Information Technology options list. You must select IAB203 if it's not in your first major.	
IAB203	Process Modelling
Information Technology unit options list.	
Process Analytics and Automation Minor Option units. Select one unit from the Process Analytics and Automation Option Units list.	
DSB201	Advanced Databases
DSB202	Data Ethics and Society
IAB201	Modelling Techniques for Information Systems
IAB204	Business Analysis for IT Systems
IAB353	Data Analytics for Enterprise Systems

Software Development Minor	
Code	Title
IN01MNR-SOFDEV v1	
Software Development Minor Option Units. Complete 48 credit points of study (four units) selected from the Software Development Minor Option Units list	
CAB201	Object-Oriented Programming and Design
CAB230	Web Computing
CAB301	Algorithms and Complexity
CAB302	Agile Software Engineering
CAB432	Cloud Computing
IAB251	Software Engineering for Enterprise Systems

Science, Information Technology and Mathematics Undergraduate University Wide Electives

Rules for selecting Unit Options:

- you must obey any Unit Option rules as set out in your course requirements
- you cannot select a unit that forms part of the compulsory units of your course or the compulsory units of your chosen major area
- you must have successfully completed any applicable pre/co-requisite units
- the offering of these units is subject to sufficient student enrolment numbers and staff availability
- University-wide Elective unit lists can be viewed via your Study Plan in eStudent.

These unit offerings are current at the time of publication but are subject to change.

In this list

Information Technology Unit Options List	
Code	Title

Computer Science	
Code	Title
CAB201	Object-Oriented Programming and Design
CAB210	User Experience Fundamentals
CAB222	Networks
CAB230	Web Computing
CAB301	Algorithms and Complexity
CAB302	Agile Software Engineering
CAB310	Interaction and Experience Design
CAB330	Machine Learning for Decision Making
CAB340	Cryptography
CAB401	High Performance and Parallel Computing
CAB403	Systems Programming
CAB432	Cloud Computing
CAB440	Network and Systems Administration

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Computer Science	
CAB443	Systems Security
CAB444	Secure Network Architectures
DSB100	Fundamentals of Data Science
IFB102	Introduction to Computer Systems
IFB104	Introduction to Programming
IFB240	Cyber Security
IFB320	Generative AI
IFB343	Secure Software Development
IGB120	Introduction to Game Design
IGB180	Games History, Culture and Psychology
IGB283	Game Engine Theory and Application
IGB388	Design and Development of Immersive Environments

Information Systems	
Code	Title
DSB201	Advanced Databases
DSB202	Data Ethics and Society
IAB201	Modelling Techniques for Information Systems
IAB203	Process Modelling
IAB204	Business Analysis for IT Systems
IAB207	Rapid Web Application Development
IAB230	Design of Enterprise IoT
IAB246	Organisations and Security: Governance, Risk and Compliance
IAB251	Software Engineering for Enterprise Systems
IAB260	Social Technologies
IAB305	IT Strategy and Management
IAB320	Process Improvement
IAB321	Process Technologies
IAB330	Applied Internet of Things
IAB353	Data Analytics for Enterprise Systems
IAB401	Enterprise Architecture
IAB402	IT Consulting and Leadership
IAB410	Enterprise Data & AI Governance
IFB103	IT Systems Design
IFB105	Database Management
IFB201	Introduction to Enterprise Computing
IFB220	Introduction to AI for IT Professionals
IFB452	Blockchain Technology

Science and Mathematics Unit Options List	
Code	Title

Biology and Environmental Science	
Code	Title
BVB101	Foundations of Biology
BVB103	Evolution and the Diversity of Life
BVB201	Biological Processes
BVB202	Experimental Design and Quantitative Methods
BVB202	Experimental Design and Quantitative Methods
BVB204	Ecological Science

Biology and Environmental Science	
BVB225	Ecosystems and Biodiversity
BVB305	Microbiology and the Environment
BVB306	Plant Biology
BVB313	Molecular Ecology
BVB321	Ecosystem Protection
BVB328	Applications in Biotechnology
BVB330	Synthetic Biology

Chemistry and Physics	
Code	Title
PVB210	Stellar Astrophysics
PVB220	Cosmology

Earth and Atmospheric Sciences	
Code	Title
CLB100	Global Change
CLB221	Introduction to Climate Change
CLB222	Oceans and Atmosphere
CLB333	Dynamic Atmosphere

Mathematical Sciences	
Code	Title
DSB101	Introduction to Data Science and Visualisation
DSB102	Introduction to Machine Learning
DSB200	Applied Data Science
DSB301	Advanced Visualisation and Data Science
MXB100	Introductory Calculus and Algebra
MXB103	Introductory Computational Mathematics
MXB105	Calculus and Differential Equations
MXB106	Linear Algebra
MXB106	Linear Algebra
MXB107	Probability and Statistics
MXB161	Computational Explorations
MXB201	Advanced Linear Algebra
MXB202	Advanced Calculus
MXB225	Modelling with Differential Equations 1
MXB226	Computational Methods 1
MXB241	Probability and Stochastic Modelling 2
MXB242	Regression and Design
MXB261	Game Theory and Simulation for Decision-Making
MXB322	Partial Differential Equations
MXB325	Modelling with Differential Equations 2
MXB326	Computational Methods 2
MXB332	Optimisation Modelling
MXB334	Operations Research for Stochastic Processes
MXB341	Statistical Inference
MXB344	Generalised Linear Models
MZB101	Modelling with Introductory Calculus

Faculty of Science	
Code	Title
SCB300	Professional Practice

Unit Synopses

BVB101 Foundations of Biology

Equivalents	BZB210
Credit Points	12

Biology is the study of living things. Cells are considered the basic structural unit of life, existing in diverse forms from simple single-celled microbes to complex multicellular organisms such as plants and animals. Using collaborative approaches in workshops and the laboratory you will investigate the diverse nature of cells and consider how they are built and powered and how they interact and reproduce and form whole organisms. You will use the concepts developed in this unit to discuss more complex questions such as “are viruses alive” and “can we synthesise life”. This unit has a strong focus on development of practical skills and engagement in these activities is required to be able to undertake assessment that supports these learning outcomes.

View unit details online (current students only)
View unit timetable

BVB103 Evolution and the Diversity of Life

Equivalents	BVB102
Credit Points	12

The famous biologist, Theodosius Dobzhansky (1973) stated that "nothing in biology makes sense except in the light of evolution". Studying evolution and biodiversity provides you with the fundamental basis for understanding biological patterns and processes from data sources spanning research from genomes to ecology and fossils. This unit provides an overview of the nature of inheritance, biological variation, natural & sexual selection and adaptation as well as the diversity of life that these processes have generated. The unit aims to equip you to be able to (1) interpret any biological observation in an evolutionary framework in fields as diverse as conservation and medical genetics and (2) recognise and classify major groups of organisms. We will introduce some key elements of Australia’s largely unique flora and fauna. As a foundational unit, knowledge and skills that you gain will be built upon in subsequent units in the biology and environmental science majors.

View unit details online (current students only)
View unit timetable

BVB201 Biological Processes

Pre-requisites	BVB101 or BZB210 or ((BVB203 or BVB306) and admission to ST20)
Credit Points	12

An understanding of processes which occur at the cellular level is fundamental to all aspects of biology. Using a combination of theoretical and laboratory-based approaches to enquiry you will explore the biochemical pathways and processes that facilitate biological function and the genetic mechanisms that control them. You will investigate how this information can be applied to solve current biological problems.

View unit details online (current students only)
View unit timetable

BVB202 Experimental Design and Quantitative Methods

Pre-requisites	SEB113 or MAB101 or MAB141 or MXB101 or MXB107 or (MZB103 and (MZB104 or MZB105))
Credit Points	12

The skills to design, carry out, analyse and interpret experiments are fundamental for scientists. This is particularly important for students of biology and environmental science since environmental variability is an inherent element of the systems that they will need to understand. This unit builds on first year knowledge of the theory of science and the scientific method developed in first year. With field trips it provides real world experience in problem assessment, formulation of testable hypotheses, and experimental design. These critical skills are used and further developed in later units.

View unit details online (current students only)
View unit timetable

BVB204 Ecological Science

Pre-requisites	BVB101 or BVB102 or BVB103 or EVB102 or (admission to ST20 and completion of 96 credit points)
Credit Points	12

Ecology is the study of the living and non-living factors that influence the distribution and abundance of organisms. It is a key discipline of biological and environmental science, and is central to managing and conserving species, ecosystems and to the short and long-term sustainable management of natural resources. This unit examines the fundamental concepts of ecology such as population ecology, interactions between trophic levels in food webs, attributes of ecosystems such as biodiversity and nutrient cycling; and develops the conceptual foundation for later subjects in the biological and environmental science majors and minors.

View unit details online (current students only)
View unit timetable

BVB225 Ecosystems and Biodiversity

Pre-requisites	BVB101 or BVB102 or BVB103 or EVB102 or BZB210 or admission to ST20
Anti-requisites	BVB223, BVB224
Credit Points	12

There are over one million species catalogued, and several million yet to be described. Familiarity with biological diversity and the role species play within ecosystems is central to biological and evolutionary research, and for developing solutions to sustain the environmental health of our planet. Australia has a largely unique flora and fauna, with a very high proportion of our plant and animal species found nowhere else. While it is impossible to cover every Australian plant and animal in a single unit, we will introduce some key elements of that diversity, with a focus on terrestrial ecosystems and the plants, insects and mammals which occur in them. For students enrolled in biology and environmental science majors this unit provides foundational knowledge to complement areas such as ecology and environmental monitoring; while for education majors the skills gained in creating collections forms a basis for teaching biology in both primary and secondary school.

View unit details online (current students only)
View unit timetable

BVB305 Microbiology and the Environment

Pre-requisites	BVB201
Credit Points	12

The biology, evolution and ecology of microorganisms underpin critical, complex and, sometimes, harmful processes in almost every environment on earth, from oceanic vents to the animal gut, in the roots of plants and the health of soil, and even in your shower head. These fundamental biological processes are a rich resource for cutting-edge research and applications of biotechnology, including the growing field of synthetic biology. You will conduct original laboratory research and desktop analyses to explore, evaluate and communicate insights into core concepts in microbiology, their functional interactions in the environment, and their exploitation in agriculture, bioindustry and the environment. Through this, you will build skills in project planning, design, conduct, analysis and communication to address real world applications and prepare you for a key area of post-graduate employment.

View unit details online (current students only)
View unit timetable

BVB306 Plant Biology

Pre-requisites	BVB201 or admission to ST20 or (BVB101 and BVB103 and admission to SV02)
Equivalents	BVB203
Credit Points	12

This is a foundational plant biology unit addressing the core concepts of plant function, including structural, physiological and molecular aspects, from the cell to the whole plant. This subject will outline and teach the skills required for measuring and monitoring plant function and an appreciation of how they are influenced by the environment and applied by industry. It introduces the fundamentals of plant physiology, biochemistry and molecular biology in such a way to enable you to understand how plants grow, develop and interact with their environment, and will also be valuable for lifelong appreciation of the potential of agriculture and its contribution to industry and humanity.

View unit details online (current students only)
View unit timetable

BVB313 Molecular Ecology

Pre-requisites	BVB204 or (BVB201 and admission to ST20) or (BVB101 and BVB103 and admission to SV02)
Credit Points	12

Understanding the dispersal and movement of genes in populations is fundamental to the management of invasive species, the management of wild resources and the conservation of rare species. In this unit, you will synthesize scientific literature, develop and apply problem solving skills through genetic data analysis and critical evaluation of results in molecular ecological research, and communicate outcomes in written format. This unit will provide the theoretical and analytical training required for practicing ecologists to use genetic techniques in research and applied settings.

View unit details online (current students only)
View unit timetable

BVB321 Ecosystem Protection

Pre-requisites	BVB204
Credit Points	12

Invasive species cause substantial and costly negative effects to native ecosystems. An understanding of the ecological processes by which they are introduced, establish and spread in new regions is essential for their control and to ensure the protection of ecosystems. Invasive species are now so widespread that they will be encountered in some way in a wide range of careers in ecology and environmental science. Invasive species can cause devastatiing effects in native ecosystemsand within agro-ecosystems. Managing invasive species populations in a sustainable manner requires science and sophistication. Integrating and extending work introduced in earlier units, you will learn the skills and concepts that are necessary to understand, analyse and manage invasive species, and the processes of biological invasion.

View unit details online (current students only)
View unit timetable

BVB328 Applications in Biotechnology

Pre-requisites	BVB317 or BVB201
Credit Points	12

Biotechnology underpins the manufacture of a wide variety of products on which society depends. Such products include pharmaceuticals, chemicals, fuels, and foods. With the major advances in biotechnology and the ability to engineer biological systems in recent decades, biotechnology offers the generation of better products at lower cost with reduced environmental impacts. As such, both established companies and start-ups are making a major impact on the global stage to deliver products that we use every day. You will learn about a variety of biotechnology applications from both a technical and a business perspective. Your learning will be contextualised with local and international biotechnology company case studies and you will perform an entire biotechnology process during practical classes. At the end of the unit you will have a deep understanding of the biotechnology industry and the technologies that underpin it as well as an awareness of the opportunities it offers.

View unit details online (current students only)
View unit timetable

BVB330 Synthetic Biology

Pre-requisites	BVB317
Anti-requisites	BVB314, BVB327
Credit Points	12

The field of Synthetic Biology has emerged from decades of advances in DNA sequencing and editing technologies. Thanks to the increasing information available describing DNA sequence-function relationships, scientists can design and build new biological parts (like enzymes), new genetic programs (connecting multiple biological parts), and even new organisms. These products of synthetic biology are used improve agricultural practices, reduce pollution, develop new therapeutics, and create new bio-based chemicals and materials. In this unit you will be introduced to bioinformatic tools for analysing DNA, and for designing new genetic circuits. You will apply this knowledge to design, edit and implement genetic programs *in vitro* and in live organisms, and develop data analysis skills.

View unit details online (current students only)
View unit timetable

CAB201 Object-Oriented Programming and Design

Pre-requisites	IFB104 or ITD104 or MZB126 or MZB127 or EGD126 or MXB103 or EGB103 or EGD103
Equivalents	CAZ201, ITD121
Credit Points	12

This unit builds on foundational programming skills introduced in IFB104, EGB103, or MZB126, where you learned to create algorithms using sequence, selection, and iteration and applied functions for code abstraction and reuse. In this unit, these core concepts are reinforced and extended with additional abstraction techniques that help manage complexity in larger software systems. You will also be introduced to object-oriented principles that organise solutions around classes - a real world representation of objects. This approach achieves a flexible architecture to evolve to changing user needs. This unit prepares you for more advanced programming units.

View unit details online (current students only)
View unit timetable

CAB210 User Experience Fundamentals

Pre-requisites	IFB103 or ITD103 or EGB101 or IGB120
Credit Points	12

User experience (UX) means how a person feels when interacting with digital technology, like mobile applications, web services and games. This unit introduces user experience methods to study people's needs in a real-world context, and to evaluate the usability and experience with technologies. This unit is important to inform the design and development of technologies that meet the needs of people who are going to use them and the context within which they will be used. A stronger understanding of user experience will provide you with an edge in the market place for jobs such as interaction designers, usability engineers, game designers, app developers, information architects, and user experience designers.

View unit details online (current students only)
View unit timetable

CAB222 Networks

Pre-requisites	IFB102 or ITD102 or EGB120
Equivalents	CAB303
Credit Points	12

Building on your digital systems knowledge, you will be introduced to practical and theoretical knowledge on a wide range of modern networking topics to be able to design, implement and maintain network-based applications. You will participate in practical networking exercises to provide hands-on experience with network-based computing. You will gain proficiency in using network tools commonly used by industry professionals for traffic capture and analysis, security analysis, and troubleshooting in production networks —essential skills required for network operations in the IT industry.

View unit details online (current students only)
View unit timetable

CAB230 Web Computing

Pre-requisites	IFB104 or ITD104 or EGB103 or EGD103
Equivalents	INB271
Credit Points	12

The World Wide Web is the most important platform for software systems and an integral part of modern life. Many companies owe their existence to the web, through applications deployed over the Internet using web protocols. All IT professionals require a good understanding of the web and its architecture, especially software developers and those tasked with maintaining and implementing web-based software systems. This unit is a technical introduction to modern web computing. You will design and implement clean and responsive user interfaces, taking account of accessibility and internationalisation. We will provide an introduction to JavaScript and you will use it throughout the semester, gaining practical experience with HTML, CSS and frameworks such as React on the client side, and node.js, Express and the node ecosystem on the server side. You will explore security threats and their mitigation and gain practical experience deploying an internet facing web server using HTTPS.

View unit details online (current students only)
View unit timetable

CAB301 Algorithms and Complexity

Pre-requisites	CAB201 or ITD121
Equivalents	CAZ301
Credit Points	12

This unit teaches you classical data structures and algorithms used in software development and the fundamental principles for analysing the time efficiency of algorithms, allowing you to distinguish software solutions that can process large amounts of data or perform complex calculations effectively from those that run unacceptably slowly or not at all. In this unit you will examine a range of different algorithms, review the principles used to predict their efficiency and perform empirical measurements of specific algorithms to confirm the theoretical predictions.

View unit details online (current students only)
View unit timetable

CAB302 Agile Software Engineering

Pre-requisites	CAB201
Equivalents	CAZ302
Credit Points	12

This unit teaches you how to work effectively in a team to develop large-scale software systems. It includes principles of teamwork, modern software development methodologies, agile project management, and tools that are needed when working in a team on a large project. You will learn how to design and implement an accessible and usable application, testing the code quality through industry-relevant practices and methodologies. You will learn how to work and thrive in a diverse and interdisciplinary team, and will work on industry-inspired projects, receiving feedback and support to your learning throughout the project journey.

View unit details online (current students only)
View unit timetable

CAB310 Interaction and Experience Design

Pre-requisites	CAB210
Credit Points	12

This is an advanced human-computer interaction unit focussed on the design and evaluation of emerging technologies for human use. Emerging technologies like robots, machine learning, and Internet of Things devices have the potential to disrupt how we work and live. Based on interaction design theories and methods, this unit focusses on how we design, prototype, and study how people experience such emerging technologies. Understanding how to design emerging technologies for human use will give you an edge in the market place for jobs as user experience professionals, IT developers, and interaction designers. The theories and methods introduced in this unit prepare you for a career in human-computer interaction research and development. This unit builds on design thinking skills from unit IFB103 and user experience research skills from unit CAB210 to design experiences with emerging technologies.

View unit details online (current students only)
View unit timetable

CAB330 Machine Learning for Decision Making

Pre-requisites	DSB100 or DSB102 or CAB201 or IFB220
Credit Points	12

Machine Learning has become a popular technology for decision-making, transforming an organisation's large data collection into actionable insights about customers and business processes. It has direct applications in several fields, including social networks, business processes, search engines, e-commerce, digital libraries, bioinformatics and web information systems. This unit provides fundamental knowledge and skills in data analytics and machine learning to support data-driven decision-making in diverse and interdisciplinary applications. You will learn about machine learning and data mining techniques, including classification, clustering and association mining, and explore how these AI methods can be applied to text and web usage data. This is an introductory unit, and the knowledge and skills developed here are relevant to all IT professionals.

View unit details online (current students only)
View unit timetable

CAB340 Cryptography

Pre-requisites	IFB240
Credit Points	12

The dependence of modern society on remote electronic data transmission and storage makes it an essential requirement that this data be secured, both against unwanted disclosure and malicious alterations. This unit provides a self-contained introduction to the field of cryptography, from historical roots and attacks, to the mathematical principles that underpin the workings of the modern ciphers most commonly in use for securing internet communications. The focus of this unit is on a grounded understanding of cryptographic designs and their limitations, which in turn inform how they are used in practice.

View unit details online (current students only)
View unit timetable

CAB401 High Performance and Parallel Computing

Pre-requisites	IFN584 or IFQ584 or (IFN563 and IFN564) or (IFQ563 and IFQ564) or CAB301
Credit Points	12

Building on your skills in "sequential" programming, this unit teaches you the tools and techniques needed to exploit multi-processor computer systems to achieve dramatic performance improvements for computationally intensive problems. This unit gives you both an understanding of why future computer hardware will be increasingly parallel, the challenges this poses for software development as well as a set of practical skills in creating high-performance programs using today's best tools and techniques.

View unit details online (current students only)
View unit timetable

CAB403 Systems Programming

Pre-requisites	((CAB201 or ITD121) and CAB222) or CAB202 or EGB202
Credit Points	12

Most of our other Computer Science units focus on high-level applications programming. Applications programmers are insulated from the low- level intricacies of the underlying hardware by making use of services provided by the operating system such as threads, virtual memory, file systems and device drivers. This unit focuses on Systems Programming, where the programmer can't necessarily rely on high level services provided by the operating system and must interact directly with the underlying hardware. Systems software is either part of the operating system or software that operates at a similar level. This unit aims to give you practical programming skills for developing systems level applications and services.

View unit details online (current students only)
View unit timetable

CAB420 Machine Learning

Pre-requisites	(CAB201 or EGB202 or CAB202 or ITD121 or IFN501 or IFN556 or Admission to (EN50 or EN55 or EN52 or EN56 or EN57 or EN62 or EN72)) or (192cps in SV03 or IV04 or MV05 or EV08) or (144cps in EV10) or (enrolment in IV53 or IV54 or IV55 or IV56 or IV58).
Anti-requisites	IFN580

Pre-requisites	(CAB201 or EGB202 or CAB202 or ITD121 or IFN501 or IFN556 or Admission to (EN50 or EN55 or EN52 or EN56 or EN57 or EN62 or EN72)) or (192cps in SV03 or IV04 or MV05 or EV08) or (144cps in EV10) or (enrolment in IV53 or IV54 or IV55 or IV56 or IV58).
Credit Points	12

Machine learning is the science of getting computers to act without being explicitly programmed. This unit provides you with a broad introduction to machine learning and its statistical foundations. Topics include: definition of machine learning tasks; classification principles and methods; dimensionality reduction/subspace methods; and deep learning methods such as convolutional neural networks and transformers. The unit makes use of python, jupyterlab, git and state of the art machine learning libraries. In addition to addressing specific machine learning methods, we will consider the ethical implications of machine learning in applications where individuals or groups could be marginalised, and the computational cost of machine learning methods and ways to reduce the compute burden. Application examples are taken from areas such as computer vision, finance, market prediction and information retrieval.

View unit details online (current students only)
View unit timetable

CAB432 Cloud Computing

Pre-requisites	CAB301 or CAB302 or INB370 or INB371 or IFN666 or IFQ666 or (IFN582 and IFN584) or (IFQ582 and IFQ584)
Equivalents	CAZ432
Credit Points	12

Cloud Computing is among the most important developments in the IT industry in recent years, and one which has received enormous attention. Cloud is a natural progression from earlier trends in service and infrastructure outsourcing and virtualisation, but is distinguished by its elasticity and scale: service and infrastructure provisioning may change rapidly in response to variations in demand, allowing clients to cater for unexpected spikes in load without tying up capital in expensive and potentially underutilised assets. Cloud services and technologies are becoming increasingly diverse and sophisticated, moving rapidly from the original 'bare metal' offerings and providing a rich set of options and APIs. This unit provides a technically oriented introduction to Cloud Computing, giving you experience in developing modern cloud applications and deploying them to the public clouds of the major vendors.

View unit details online (current students only)
View unit timetable

CAB440 Network and Systems Administration

Pre-requisites	CAB222 or CAB303 or INB251
Credit Points	12

This unit builds on your existing knowledge of networks and systems administration. You will be introduced to technical knowledge and practical skills for managing network administration, including: (1) configuring addressing and routing with physical/virtual network devices, (2) installing/maintaining/troubleshooting network services on a Unix-like platform, and (3) preventing vulnerabilities/threats to network systems and proposing mitigation strategies to secure network infrastructure. CAB441 Network Security builds on this unit by extending your network administration skills to secure network application services.

View unit details online (current students only)
View unit timetable

CAB443 Systems Security

Pre-requisites	IFB103 and IAB246 (IAB246 can be studied in the same semester)
Credit Points	12

Systems security focuses on the organisational practices that ensure the security requirements of systems, which are composed of interconnected components and connections and the networking software that supports those interconnections. More specifically, systems security requires a holistic design thinking approach to systems. This includes the importance of security policy models, as well as organised identification and authentication processes. Systems security also addresses system access control, system management, system control processes, system recovery and system security documentation.

View unit details online (current students only)
View unit timetable

CAB444 Secure Network Architectures

Pre-requisites	CAB222 and (CAB440 or CAB443)
Credit Points	12

As a system administrator or information security professional, you are expected to have an in-depth understanding of various network security controls, principles, and analysis tools and their use in a wider cybersecurity context. In this unit, you will analyse real-world security incidents and learn to use tools and protection measures to design, build, test and maintain an organisation's security infrastructure that can effectively withstand cyber threats and fortify defenses.

View unit details online (current students only)
View unit timetable

CLB100 Global Change

Equivalents	ERB211
Credit Points	12

In CLB100, you will discover how the Earth has undergone natural global change and how this compares to anthropogenic change of the planet. The three key aims of the unit are: i) to focus on climatic and global change of the last 3 million years; ii) to then put this into the context of the Earth's long planetary history; and finally, iii) to develop a sound appreciation of the close relationship between human evolution and global change. You will gain new appreciation of diverse perspectives and inclusion by learning about how different cultures have recorded past global change. In conclusion, the unit will let your discover how past planetary change can be used to inform models predicting future climate: the past is the key to the future.

View unit details online (current students only)
View unit timetable

CLB221 Introduction to Climate Change

Equivalents	PQB360
Credit Points	12

This unit is designed to offer science, engineering and other students an opportunity to understand fundamentals of climate and climate change together with sustainable development efforts related to clean energy technologies. It provides students with an overview of global climate and climate change drivers, meteorological parameters and global air circulation, as well as an overview of technological pathways towards low carbon society. Students will explore global energy balance and climate change through an investigation of (i) Energy related environmental problems on local and global scale; (ii) Earth's climate, meteorology and transport of pollutants in the atmosphere; (iii) Working principles in selected conventional and alternative energy technologies to reduce energy related environmental consequences.

View unit details online (current students only)
View unit timetable

CLB222 Oceans and Atmosphere

Equivalents	ERB202
Credit Points	12

Oceans make up 70% of the Earth's surface, yet less than 5% of them have been explored. There is therefore still much to learn about the marine environment, marine resources, and management, and how oceans affect atmospheric circulation. This unit takes a bottom-up approach introducing students to the major geological processes and geomorphology features that shape the ocean floor, dynamic sediments, and biology that are sourced and distributed by ocean currents and chemistry, followed by an overview of the factors governing ocean circulation and ocean water properties and finally looking at how oceans affect atmospheric circulation and climate. The interaction of all these processes has a direct societal impact such as the management of marine resources, including the cultural narratives of the Great Barrier Reef, infrastructure and food security, and risk mitigation of natural hazards, weather patterns, and climate change.

View unit details online (current students only)
View unit timetable

CLB333 Dynamic Atmosphere

Pre-requisites	PQB360 or CLB221
Equivalents	ERB311
Credit Points	12

Building on the foundation of atmospheric science laid in PQB360, this unit will provide enhanced understanding of chemical and physical processes that are responsible for structure, composition and properties of the atmosphere. Students will gain deeper understanding of aerosol and cloud formation, their interaction with solar radiation and role in global climate, as well as chemical transformations that govern abundance of important atmospheric trace species in both gas and aerosol phase. This knowledge will provide students with the background to understand current issues, such as stratospheric ozone depletion, impact of volcanoes on climate, air pollution and photochemical smog, acid rain and climate change.

View unit details online (current students only)
View unit timetable

DSB100 Fundamentals of Data Science

Pre-requisites	(CAB201 or ITD121) and (MZB151 or any MXB unit). CAB201 can be studied in the same teaching period as DSB100.
Credit Points	12

Data is becoming central to every organization's decision making process, and the demand for data savvy modelers and software engineers is rapidly increasing. Modern computational approaches to data analysis have to enable users to acquire, manage, interpret, present and disseminate large volumes of heterogeneous data. Data science is a synthesis of statistics, mathematics, machine learning and computer science, and uses tools, techniques, and approaches from all of these fields to extract information from datasets. This unit will introduce you to a wide range of Data Science methods and theories to model and analyze data.

View unit details online (current students only)
View unit timetable

DSB101 Introduction to Data Science and Visualisation

Equivalents	MXB262
Credit Points	12

Our world has an unprecedented amount of available data - especially in STEM, where generating and working with data is core to our fields. The ability to analyse and visualise data is critical for exploring and communicating science and engineering findings. Modern data science and visualisation techniques allow us to efficiently explore and communicate data.

View unit details online (current students only)
View unit timetable

DSB102 Introduction to Machine Learning

Credit Points	12

This unit introduces you to foundational concepts in statistical machine learning, equipping them with essential skills to handle and analyse complex data. You will explore both supervised and unsupervised learning techniques, starting with linear regression and advancing to methods like decision trees, support vector regression, and introductory neural networks. Additionally, the unit covers essential clustering techniques and simple yet practical machine learning applications suitable for first-year data science students. Through a combination of lectures, tutorials, and both individual and group assignments, you will engage deeply with real-world problems, and have the opportunity to benefit from diverse perspectives and career supports to develop their employability. You will be prepared to apply these methods and use industry-relevant digital practices to a range of real-world data problems and lay the groundwork for advanced studies in data science.

View unit details online (current students only)
View unit timetable

DSB200 Applied Data Science

Pre-requisites	DSB101 and DSB102 and IFB104
Credit Points	12

Thanks to information technology, *data* has become the life blood of human endeavour. Individually and collectively, we depend on digital data to live well and flourish. This is because (even though data are merely symbols that can carry or store information) we can turn data into information, knowledge and wisdom that we can preserve, share and apply. *Data Science* is all about making sense of the information that data may hold to help us understand our Universe and act wisely to deliver benefit and avoid harm. *Applied data science* is where we bring abstract concepts, theories, methods and algorithms to bear on real-world data to inform human decisions and actions.

View unit details online (current students only)
View unit timetable

DSB201 Advanced Databases

Pre-requisites	(IFB104 or ITD104 or EGB103) and (IFB105 or IFB130 or ITD105)
Equivalents	IAB206
Credit Points	12

This unit introduces you to the technologies that can be used to address challenges in managing fast incoming, voluminous, and varied data that is increasingly being relied on to make decisions in today's business environment. You will develop practical skills in using advance database technologies that will prepare you to be a data analyst, business analyst, solution architect, as well as enterprise architect.

View unit details online (current students only)
View unit timetable

DSB202 Data Ethics and Society

Pre-requisites	IFB104 and IFB105
Credit Points	12

In this unit you will explore ethical concerns associated with information and data, and develop a personal ethical stance which will help you approach data science tasks in an ethical way. The unit will investigate ethical problems that can arise in the use of information including issues associated with fairness and dignity, transparency and privacy, sovereignty and the law, governance and regulation, AI and emerging technologies, and balancing competing rights. The unit will use real examples from contemporary applications of data science and information technologies, analysing the practical effects of good versus poor ethical practices. You will use ethical theory to help identify your own personal stance on ethical issues, and in doing so shape your own ethical position and provide you with approaches that you can use in practice. Successful completion of this unit will prepare you to be able to face ethical issues in your future work and respond thoughtfully to them.

View unit details online (current students only)
View unit timetable

DSB301 Advanced Visualisation and Data Science

Pre-requisites	DSB102 and DSB200
Credit Points	12

With the proliferation of sensor and computational technologies, modern scientists and decision makers are faced with increasingly complex datasets and problems. Referred to as high dimensional data, these contexts are characterised by large numbers of variables and complex interactions and relationships between them. The tools and techniques for modelling, analysing, making predictions of and visualising this data underpin many disciplines. These include the weather, genetics, health, sports, the environment, engineering systems and financial systems.

View unit details online (current students only)
View unit timetable

DXB212 Tangible Interaction Design

Pre-requisites	DXB211 or IFB104
Credit Points	12

This unit provides an in-depth exploration of tangible interaction design through the creation of an advanced prototyping project. It integrates theoretical frameworks with hands-on practice, focusing on the design of computational and interactive prototypes that engage users through tangible and embodied interaction. You will develop proficiency in rapid prototyping techniques, producing both low-fidelity and high-fidelity prototypes, using a variety of design and fabrication methods.

View unit details online (current students only)
View unit timetable

IAB201 Modelling Techniques for Information Systems

Pre-requisites	((IFB103 or ITD103) and (IFB105 or IFB130 or ITD105)) or IAB203
Equivalents	ITD122
Credit Points	12

This is an introductory unit that will provide you with the foundational skills and knowledge required for understanding, designing and analysing information systems. The unit aims to develop an ability to manage the complexity of contemporary and future information systems and the domains in which they are used. It will also provide you with the skills to design artefacts, fit for purpose and audience, that can be used to solve real-world problems related to information systems. Unit content will play an important role in future units and a wide variety of professional IT activities. This unit expands on knowledge acquired in IFB103: IT Systems Design by introducing conceptual modelling techniques that underpin most modern systems modelling languages. Subsequent units will build on the conceptual modelling skills learned in this unit, for example, by applying it to the techniques covered in IAB203: Business Process Modelling and IAB204: Business Requirements Analysis.

View unit details online (current students only)
View unit timetable

IAB203 Process Modelling

Credit Points	12

This unit introduces business process management concepts: how organisations improve their processes in terms of time, cost and quality. It introduces process identification and process discovery. Furthermore, it addresses the fundamentals of process modelling: model quality, correctness issues and modelling in BPMN's collaboration and choreography diagrams. After this unit, IAB320 continues with other business process improvement steps.

View unit details online (current students only)
View unit timetable

IAB204 Business Analysis for IT Systems

Pre-requisites	IFB103
Credit Points	12

This unit introduces you to the role, knowledge, skills and techniques required of a business analyst. The unit focuses on the tools and methods used by a business analyst, as well as the soft skills such as creativity and communication, all of which are critical to successful business requirements analysis.

View unit details online (current students only)
View unit timetable

IAB207 Rapid Web Application Development

Pre-requisites	(IFB103 or ITD103) and (IFB105 or ITD105 or IFB130)
Equivalents	IAZ207
Credit Points	12

Web applications are widely in use both within enterprises and in consumer applications. Developing Web applications faces significant challenges, including faster delivery of new innovations, robustness for change, and performance scalability. The unit will address these challenges by using Model-View-Controller (MVC) frameworks to support rapid development of web applications. The knowledge and skills developed in this unit are valuable for all IT professional roles – software engineers, business analyst and architects, enabling an understanding of software systems design practices and development practices. This is an introductory unit and you will be exposed to web application development through a guided process of using well known frameworks such as CSS-Bootstrap, Python-Flask and JQuery. It builds on concepts learnt in IFB103 and IFB105 and recommends knowledge of Python programming.

View unit details online (current students only)
View unit timetable

IAB230 Design of Enterprise IoT

Pre-requisites	IFB104 or ITD104 or EGB103
Credit Points	12

This unit introduces the components of a mobile ubiquitous system, including stand alone and wearable sensors and wireless network protocols. It introduces the Internet of Things (IoT) context and develops the skills in designing systems and applications that use mobile and ubiquitous sensors and smart devices. The ability to critically review real case studies, expand awareness of interconnections between technologies, networks and user contexts and design a solution to a smart IT context problem is a requirement for a range of graduate positions. This is the first unit in the IoT minor and builds on the skills that you developed in IFB103 IT Systems Design, and IFB104 Building IT Systems. IAB330 Applied Internet of Things builds on this unit in which you design and build a working prototype system that uses mobile and ubiquitous system components.

View unit details online (current students only)
View unit timetable

IAB246 Organisations and Security: Governance, Risk and Compliance

Pre-requisites	IFB240 (can be studied concurrently)
Credit Points	12

Cybersecurity is the practice of safeguarding an organisation's critical infrastructure from cyber threats, including ransomware, malware, and phishing. Critical infrastructure encompasses IT components (software, platforms, computing infrastructure, networks, and other assets) and physical spaces. Infrastructure attacks have increased the need for cyber risk management skills. This unit introduces cybersecurity and governance frameworks, covering security practices across people, processes, and technology. Topics include threat assessment, risk management, incident response, security compliance, and policy management. From real-world case studies and problems, you will learn to prepare for roles as cybersecurity professionals, gaining theoretical knowledge and practical skills to prevent and respond to cyberattacks. This unit also prepares you for advanced studies in the cybersecurity major.

View unit details online (current students only)
View unit timetable

IAB251 Software Engineering for Enterprise Systems

Pre-requisites	((IFB104 or ITD104 or EGB103) and IFB201) or CAB201
Equivalents	IAZ251
Credit Points	12

This unit provides knowledge and skills related to software engineering, applied to the design, development, configuration, and maintenance of Enterprise Systems. Enterprise Systems (ES) are large-scale software applications that serve as the operations of modern organizations, such as managing customers, controlling production, and connecting to suppliers, and more. You will learn how to analyse business needs, design software architectures, and develop, test and deploy software applications. To support this you will learn, key aspects of software engineering using a real-world ES Platform. This will include: software design patterns; multi-tier software components and rapid composition of solutions; testing; DevOps; and agile development. By the end of the unit, you will become proficient at these for real-world projects and software development.

View unit details online (current students only)
View unit timetable

IAB260 Social Technologies

Credit Points	12

This unit introduces the theoretical foundations and practical applications of social technologies, with a focus on online networks and digital communities. You will begin by analysing the structure and dynamics of social networks to understand how people connect, influence each other, and form communities in digital spaces. Using real-world data, you will learn to identify patterns, relationships, and key actors that shape online interactions. Building on these insights, the unit then shifts to designing and developing effective online communities. You will examine the critical design, behavioural, and structural features that underpin successful digital communities and learn how to apply them in practice. As social technologies become increasingly embedded in business models and the broader digital economy, this unit equips you with the skills to evaluate and build social platforms that foster meaningful engagement and purpose-driven participation.

View unit details online (current students only)
View unit timetable

IAB305 IT Strategy and Management

Pre-requisites	IFB103 or ITD103
Equivalents	IAB205, IAZ305
Credit Points	12

This unit provides the essential skills and knowledge for managing an information system across its lifecycle, spanning inception, design, implementation and production release. For systems inception, it covers the way an information system is proposed and justified, at the highest level, using a business model and business case analysis. Systems design encompasses the planning and management aspects of both business and systems architecture to support new design proposals. Systems implementation focusses on the different options and processes for purchasing off-the-shelf solutions in support of systems design. Production release covers the planning of IT infrastructure to host and run digital solutions as well as organisational change management. The unit expands on skills from IFB103 Systems Design and IAB201 Modelling Techniques for Information Systems. The skills learned in this unit will be utilised and further developed in IAB401 Enterprise Architecture.

View unit details online (current students only)
View unit timetable

IAB320 Process Improvement

Pre-requisites	IAB203
Credit Points	12

This unit fosters the development of your process analysis, improvement, and design skills. These skills and capabilities will prepare you to undertake the digital transformation challenges of today’s organisations. You will understand and apply a variety of methods, tools, techniques, and approaches for organisational-wide process improvement initiatives. You will be exposed to a robust selection of quantitative and qualitative analysis techniques as well as key process redesign paradigms used in the industry. This will involve developing your knowledge and expertise in different process improvement methodologies such as Lean, Six Sigma and Process Reengineering using a hands-on teaching approach with real-life case studies to enable authentic learning outcomes.

View unit details online (current students only)
View unit timetable

IAB321 Process Technologies

Pre-requisites	IAB203
Credit Points	12

Whether you will be a business analyst, a data analyst, a process owner, a solution architect or a software engineer, it is essential that you understand the principles and value of process automation and process intelligence in order to fully realise the benefits of digital transformation. This unit introduces the concepts of process automation and process analytics. You will learn how to develop an executable process model for automation and how to monitor its performance. You will learn how to analyse automated processes using latest technologies in data and process mining. The hands-on approach allows you to design, control and analyse automated processes using a variety of well-known process technologies.

View unit details online (current students only)
View unit timetable

IAB330 Applied Internet of Things

Pre-requisites	(IFB104 or ITD104 or EGB103 or EGD103) and (IFB102 or EGB202 or CAB202)
Equivalents	IAZ330
Credit Points	12

This unit equips you with the theoretical foundations and practical expertise needed to design and implement innovative IoT solutions for real-world challenges. You will gain a comprehensive understanding of IoT architectures, sensor integration, data processing workflows, and the application of machine learning in IoT contexts. Through a balanced approach of lectures and hands-on activities, the unit covers critical topics such as IoT system components, hardware-software integration, data collection, preprocessing, storage strategies, and machine learning techniques customised for IoT applications. By emphasising experiential learning, you will develop end-to-end IoT solutions, addressing practical problems and honing skills essential for careers in the dynamic and fast-growing IoT industry.

View unit details online (current students only)
View unit timetable

IAB353 Data Analytics for Enterprise Systems

Pre-requisites	((IFB104 or ITD104 or EGB103) and (IFB105 or IFB130 or ITD105)) or IAB251
Credit Points	12

This unit provides knowledge and skills for supporting Business Intelligence (BI) using enterprise systems. BI is a technology-driven process for analysing data and delivering actionable outcomes as part of planning and decision-making tasks undertaken by executives, managers, and workers. It involves data science and machine learning techniques and tools applied to key aspects of businesses including products, services, customers and resources. You will be exposed to the planning, modelling, reporting, and prediction structures underpinning business intelligence. To support this, you will learn, Data preparation, analysis and modelling, predictions, and visualization. In addition, you will be exposed to advanced data analytics capabilities including, real-time analytics and Internet of Things (IoT) analytics. This will be applied through a comprehensive framework that supports data Ops, data security, and governance. The unit provides a rich exposure to real-world BI platforms.

View unit details online (current students only)
View unit timetable

IAB401 Enterprise Architecture

Pre-requisites	IFB103 AND IFB105 AND (IAB251 OR IAB203)
Equivalents	IAB301, IAZ401
Credit Points	12

This unit covers Enterprise Architecture (EA) theory and practice concerning how business and IT systems are planned and designed using modelling techniques. EA focuses on organisational capability maps, which reflect what companies do, independent of business. The techniques for capturing different artefacts relevant to systems planning at business and IT levels will include business services, processes, information and resources. You will be taught to develop a multi-layered EA using state-of-the-art techniques, such as TOGAF, Archimate, and UML modelling, based on industry-relevant digital practices and technologies. Significantly, this unit extends your knowledge and skills to model and design enterprise architecture solutions, enabling you to pursue careers in EA, modelling, design and solution architecture. The unit links to and extends learning from previous units in Data and Information Management and Process Modelling.

View unit details online (current students only)
View unit timetable

IAB402 IT Consulting and Leadership

Pre-requisites	IAB204 OR (192cps in SV03 or SV04 or IV04 or IV05 or MV05 or MV06 or BV06 or BV07 or EV08 or EV07) OR (Admission to IN20 or IN27 or IN31 or IV53 or IV57 or IV54 or IV59 or IV55 or IV52 or IV56 or IV51 or IV58 or IV60)
Credit Points	12

In this unit, you will develop a practical understanding of the IT consulting sector and the management of consulting engagements in contemporary organisations. Building on skills in business requirements analysis, the unit focuses on identifying organisational problems, evaluating solution options, and translating these insights into clear and persuasive consulting proposals. The unit emphasises professional communication, client engagement, and leadership in project-based environments. You will learn how to articulate value propositions, manage stakeholder relationships, and communicate data-informed technical and organisational requirements effectively to support evidence-based decision-making. The unit also introduces the fundamentals of establishing and managing a consulting practice, preparing you to operate as a trusted advisor and effective IT professional in modern consulting-style organisations.

View unit details online (current students only)
View unit timetable

IAB410 Enterprise Data & AI Governance

Credit Points	12

In today's rapidly evolving digital landscape with new data and artificial intelligence (AI) opportunities, effective and responsible governance is essential for organisational success, ethical integrity, and regulatory compliance. The Enterprise Data and AI Governance unit equips you with the strategic insight and practical expertise to navigate the complexities of governing digital assets confidently. You will do so by leveraging governance frameworks to create value for an organisation and its stakeholders, supported by a practical blend of theory and real-world insights. You will master good practices in corporate, data, and AI governance, ensuring alignment between organisational goals and technological capabilities. This unit uniquely equips you with the comprehensive skills needed to advance enterprises in a data-driven, AI-enabled world.

View unit details online (current students only)
View unit timetable

IFB102 Introduction to Computer Systems

Equivalents	IFB112, IND102, ITD102
Credit Points	12

This is an introductory computer science unit concerning computer systems, in particular how modern computer systems work, how they are structured, and how they operate. Computer systems are ubiquitous and yet they are unlike any other man-made product or system; they appear magical and are notoriously difficult to work with and manage in projects. This unit’s goal is to demystify computer systems so you can appreciate, understand and utilise computer systems in their subsequent learning, and effectively participate in the IT industry. You will study computers, networks, operating systems and the Web. Raspberry Pi computers will be used throughout the unit and at the end you will build their own small computer system using a Raspberry Pi.

View unit details online (current students only)
View unit timetable

IFB103 IT Systems Design

Equivalents	ITD103
Credit Points	12

The success of any IT system - whether a mobile, web app, or core platform - lies in the quality of its design. Design combines creative and technical skills to develop blueprints and prototypes that guide software development. Design covers frontend aspects where users access data and complete tasks, and backend aspects, including software components, databases, and platforms. This unit exposes you to these key design aspects. You will learn design principles and techniques alongside hands-on learning applied to real-world problems. This unit will fast-track your understanding by showing how applications like Google Maps and Facebook are designed. You will work individually and in teams, and learn to communicate design proposals using informal and formal diagrams to gain buy-in and support implementation. You will gain a coherent introduction to IT systems and the key professional roles involved in IT design work (e.g., systems/business analysts and solution architects).

View unit details online (current students only)
View unit timetable

IFB104 Introduction to Programming

Anti-requisites	CAB201, EGB103, EGD103
Equivalents	INB104, ITB001, ITD104
Credit Points	12

This unit provides a hands-on introduction to computer programming for students with no prior coding experience at all. It introduces the basic principles of programming in a typical imperative language, including expressions, assignment, functions, choice and iteration. It then shows how to use Application Programming Interfaces to complete common Information Technology tasks such as querying databases, creating user interfaces, and searching for patterns in large datasets. It finally connects the technical approaches to professional and entrepreneurial contexts. The emphasis is on developing skills through practice, so the unit includes numerous coding exercises and assignments, using a simple scripting language, a code development environment and a Generative AI tool. The unit establishes a foundation for later subjects that teach large-scale software development using industrial-strength programming languages.

View unit details online (current students only)
View unit timetable

IFB105 Database Management

Equivalents	IFB130, ITD105
Credit Points	12

This is an introductory unit on database addressing the core concepts, requirements and practices of databases. It introduces conceptual data modeling to address a key area of concern of modeling structured data to build a comprehensive understanding of the data aspect of a problem. You will learn how to transform such data model into a relational database design as well as how to effectively retrieve data through SQL queries. Normalization, database security/administration, other special topics and ethical aspects related to information systems are also covered. IAB207 Rapid Web App Development, IAB303 Data Analytics for Business Insights and the Capstone units IFB398 Capstone 1 and IFB399 Capstone 2 build on this unit for data storage/retrieval and business insights. IAB206 Modern Data Management extends this unit earning to unstructured data such as graphs and documents which are also gaining popularity in the real world.

View unit details online (current students only)
View unit timetable

IFB201 Introduction to Enterprise Computing

Equivalents	ITD201, IFZ201
Credit Points	12

This unit provides an introduction to enterprise systems, the most widely used form of software technology in the business world. It will provide you with a background of the different aspects of businesses that they manage, such as resources, teams, products, services, customers and suppliers, and prominent types of enterprise systems in the market, including customer relationship management, enterprise resource planning, supply chain management systems. Based on this background, you will learn how to design, develop and effectively utilise enterprise systems from three perspectives. These are: business processes through which activities are coordinated across an organisation; software applications which provide the programming code underpinning the processes; and decision-making in which data is combined and visualised for users to determine critical outcomes.

View unit details online (current students only)
View unit timetable

IFB220 Introduction to AI for IT Professionals

Pre-requisites	IFB104 or EGB103
Equivalents	IFZ220
Credit Points	12

In this unit, you'll discover how Artificial Intelligence (AI) is reshaping our world and driving unprecedented innovation, poised to revolutionize how we live and work. We'll cover the fundamental concepts and applications of AI, with a special focus on Generative AI. You'll learn how AI systems function, how they differ from human intelligence, and how to critically evaluate AI-generated information. We'll explore practical methods for integrating AI into IT systems, examine AI's impact on various industries, and discuss ethical, social, and legal concerns. You'll investigate how businesses plan to use AI to add value and enhance productivity, and assess the impact on jobs and skills. Through a mix of theoretical knowledge and practical exercises, you'll learn to effectively apply AI for ongoing learning, writing, and professional tasks. By the end of this unit, you'll be equipped with the foundational skills necessary to leverage AI in your future IT career.

View unit details online (current students only)
View unit timetable

IFB240 Cyber Security

Equivalents	ITD240, IFZ240, CAB240
Credit Points	12

Cyber security breaches, from database hacking to malware campaigns, are increasing. The interconnectedness of information systems means the actions of individuals impact many others. This unit is important in developing an understanding of the challenges involved in protecting information assets. Fundamental information security concepts are introduced. Security goals including confidentiality, integrity, availability, authentication and non-repudiation are defined. Threats to information and vulnerabilities that could be exploited are identified. Technical and non-technical measures to provide security for information are discussed in areas including access control, cryptography, and network communications. Cyber security management standards and best practice implementation guidelines are reviewed. You can take this unit as a stand-alone course to raise your cyber security awareness, or as a pathway to cyber security units, including network security and cryptography.

View unit details online (current students only)
View unit timetable

IFB320 Generative AI

Pre-requisites	(IFB220 or DSB102) and CAB201. CAB201 can be studied in the same teaching period as IFB320.
Credit Points	12

Generative AI is one of the most significant developments in machine learning research in recent years, and one that has received enormous attention in industry, academia and among the general public. Many industries are being transformed by advancements in this space and it is the subject of significant media attention. This unit is a technically-focused jump into the world of large language models and other forms of Generative AI, through both theory and practical applications.

View unit details online (current students only)
View unit timetable

IFB343 Secure Software Development

Pre-requisites	IFB104 and IFB240
Credit Points	12

This unit gives you the opportunity to learn a comprehensive set of skills critical to the field of secure software development. You will gain a solid understanding of the fundamental principles of secure software development, including the identification and incorporation of security requirements throughout the software development lifecycle. You will explore various categories of software vulnerabilities, along with effective strategies for their prevention, detection, and mitigation to reduce exploitation risks. You will learn to apply industry-standard frameworks, guidelines, and policies that are essential for developing secure software. The course also covers advanced security testing and analysis techniques to ensure robust protection against vulnerabilities. This unit prepares you with both theoretical insights and real-world practical skills, setting a strong foundation for a career in secure software development.

View unit details online (current students only)
View unit timetable

IFB398 IT Capstone Project (Phase 1)

Pre-requisites	[(IAB251 or CAB302 or IAB320 or IAB321 or IFB343 or CAB330 or IAB305) and completion of 120cp of (IFB% or ITD% or CAB% or IAB%)] or [192cp of DS01]
Equivalents	
Credit Points	12

This unit gives you the opportunity, under close guidance appropriate to the project, to apply and extend the knowledge and skills gained in your course to date to a substantial IT project. You will have the opportunity to gain knowledge and skills required for careful planning, scope control and task management to ensure the success of a industry-oriented project. Working in a team, you will undertake critical tasks required in the early stages of an IT project, such as: initial concept development and feasibility analysis, requirements gathering and analysis, design and project planning, evaluation/playtest. These activities will culminate in the delivery of an initial version of the IT project deliverable (e.g., software prototype, game or interactive environment, design document, literature review, environmental scan, research plan, etc.) to stakeholders as a proposal for further development in the unit IT Capstone Project (Phase 2).

View unit details online (current students only)
View unit timetable

IFB399 IT Capstone Project (Phase 2)

Pre-requisites	IFB398 or CAB398 or IAB398
Equivalents	CAB399, IAB399, IFZ399
Credit Points	12

Building upon the unit IFB398 Capstone Project (Phase 1), this unit gives you the opportunity to apply, under appropriate guidance, the knowledge and skills gained in your course to date to execute the completion of a planned IT project. In this unit, you will apply your disciplinary and professional knowledge and skills to refine and extend the existing deliverable. You will use appropriate quality assurance techniques to ensure you are meeting your stakeholders' needs. You are expected to work professionally in a team to deliver a high-quality outcome to project stakeholders. The final IT product is to be delivered as a professional package that can be used directly by stakeholders and, where appropriate, published for access by the broader community.

View unit details online (current students only)
View unit timetable

IFB452 Blockchain Technology

Pre-requisites	((((IFB103 or ITD103) and (IFB104 or ITD104)) or (EGB103 or EGD103)) and (IFB240 or ITD240)) or ((IFN581 or ((IFN555 or IFQ555) and (IFN556 or IFQ556))) and (IFN583 or ((IFN551 or IFQ551) and (IFN553 or IFQ553))) and (IFN585 or ((IFN552 or IFQ552) and (IFN558 or IFQ558))))
Credit Points	12

The unit introduces you to blockchain technology, an emerging technology that underpins major cryptocurrencies and has applications in many industry sectors. It will provide you with a background of the different aspects of applications that can benefit from blockchain, such as supply chains, energy trading, and connected vehicles, and prominent blockchain platforms in use, including bitcoin, Ethereum and Hypeledger Fabric. Based on this background, you will learn how to critically assess applications for blockchain technologies, design blockchain applications, and effectively utilise advanced features of blockchain technology.

View unit details online (current students only)
View unit timetable

IGB120 Introduction to Game Design

Equivalents	INB280, IGB220
Credit Points	12

It is crucial that a game designer understands how to create a game world, the rules that govern game play and other high level design tasks, as the result of these activities can determine whether the player finds the game enjoyable or not. This unit provides an introduction to game design, by starting with high level conceptual design tasks before moving to more concrete tasks. You will develop practical and theoretical knowledge of game design issues such as: how game systems work, how to define rules for the world, and how to balance the rules to ensure enjoyable game play. At the end of the unit, students will have worked individually and in a small project team to demonstrate their understanding of these concepts. In this unit you will analyse relevant theory, create a design document, build a prototype and undertake playtesting.

View unit details online (current students only)
View unit timetable

IGB180 Games History, Culture and Psychology

Anti-requisites	INN180
Equivalents	INB180
Credit Points	12

This unit focuses on the social, cultural, sustainability and political aspects of videogames. It serves as a survey of topics useful for those interested in the study and creation of serious games, games for health, entertainment games, virtual and augmented reality, gamification and interactive environments. The units aims to help you become well played, well versed, and well read in and of games through the development of critical media skills. By the end of the semester you will be able to speak to current academic and industry trends around games, critically assess media and studies about games for veracity and reliability, leverage critical and creative thinking to express your positions on games, and converse about the history, social-cultural and sustainability impact of games.

View unit details online (current students only)
View unit timetable

IGB283 Game Engine Theory and Application

Pre-requisites	CAB201
Anti-requisites	INB381
Credit Points	12

This unit will introduce you to the mathematics for computer graphics and games. Instead of just teaching mathematics, this unit focuses on taking mathematical theory and learning to program small examples in a game engine. The core parts of any game engine are the mathematical representations and algorithms. This unit will give you a basic understanding of the mathematics behind 3D graphics and games and the ability to apply the theory to solve problems in game engine development and software development in related areas. This unit will provide you with foundation knowledge and skills for programming and using 3D game engines. It is a pre-requisite for following advanced units that will build on these skills to provide you with enough knowledge to develop your own game engine and to have a deeper insight into popular commercial engines and tools used in the course.

View unit details online (current students only)
View unit timetable

IGB388 Design and Development of Immersive Environments

Pre-requisites	IGB200
Credit Points	12

This immersive systems unit addresses the core concepts, requirements and practices of immersive system design and development and the factors that influence the subjective user quality of immersive experiences. It introduces the major considerations and techniques for laying out immersive system designs and then the software engineering principles required to implement such systems. The ability to design and implement 3D spaces and user interactions suitable to immersive systems are covered at a general level, suitable for use in many sectors of the immersive technology industry, such as education and training, interactive entertainment, amongst others.

View unit details online (current students only)
View unit timetable

MXB100 Introductory Calculus and Algebra

Anti-requisites	MAB125, MZB125
Equivalents	MAB100,MAB120,MAB180
Credit Points	12

This unit builds on high school calculus by exploring derivatives, integrals and differential equations. It also introduces the basic theory of matrices, vectors and complex numbers. The ability to apply these concepts and techniques, and express real-world problems in mathematical language, is essential in quantitative fields such as science, business and technology. This is an introductory unit, which attempts to establish foundational skills that you will extend in subsequent discipline-specific units. This unit is particularly intended for students whose mathematics preparation does not include Queensland Senior Specialist Mathematics, Mathematics C or an equivalent.

View unit details online (current students only)
View unit timetable

MXB103 Introductory Computational Mathematics

Equivalents	MAB220
Credit Points	12

Many real world phenomena are modelled by mathematical models whose solutions cannot be found analytically. To solve these problems in practice, it is necessary to develop computational methods, algorithms and computer code. This unit will introduce you to numerical methods for addressing foundational problems in computational mathematics such as solving nonlinear ordinary differential equations, finding roots of nonlinear functions, constructing interpolating polynomials of data sets, computing derivatives and integrals numerically and solving linear systems of equations. This is an introductory unit providing foundational skills in computational methods and their practical implementation using relevant computational software. This unit will be essential throughout the remaining parts of your degree. MXB226 Computational Mathematics builds on this unit by extending your computational and programming skills to more challenging problems and more sophisticated algorithms.

View unit details online (current students only)

View unit timetable

MXB105 Calculus and Differential Equations

Anti-requisites	PVB201
Equivalents	MAB121, PVB200
Credit Points	12

Calculus and differential equations are used ubiquitously throughout mathematics, statistics and operations research. In this unit, you will build upon the foundations of calculus established in high school or in earlier university mathematics study, to greatly enhance your repertoire of theory and practice in these areas. The application of calculus and differential equations in the description and modelling of real-world problems will also be considered. This unit will extend your problem-solving skills, range of knowledge and use of techniques in differential and integral calculus. These theoretical concepts and their applications will be pursued further in MXB202 Advanced Calculus.

View unit details online (current students only)
View unit timetable

MXB106 Linear Algebra

Equivalents	MAB112, MAB122, MAB132, PVB202
Credit Points	12

This is a foundational unit in linear algebra which introduces core algebraic concepts, as well as theoretical and practical tools, that will be of central importance to solving real-world problems in science and engineering by mathematical methods. Linear algebra is fundamental to most branches of mathematics, finding widespread applications in mathematical modelling, statistics, machine learning, finance, economics, information technology, operations research, and computational mathematics. This unit aims to cultivate a deep understanding of the basic mathematical structures of linear algebra, including vector spaces and linear combinations, matrix transformations, invariant subspaces and eigenvalue problems.

View unit details online (current students only)
View unit timetable

MXB107 Probability and Statistics

Credit Points	12

Probability and statistics are essential for understanding uncertainty, making informed decisions, and analysing data across diverse fields, from business and healthcare to engineering and social sciences. This unit provides foundational skills to analyse data, test hypotheses, and draw reliable conclusions - critical for success in today's data-driven world. Students will gain hands-on experience with industry-leading software, specifically R, for data analysis. This unit focuses on core concepts and practical skills, which will be developed further in *DSB102 Statistical Machine Learning*, which introduces basic regression. Students will develop a critical, ethical approach to data, with attention to diverse perspectives. This unit also prepares students for specialised topics in later units, such as *MXB241 Stochastic Processes* and *MXB242 Regression and Design*, building skills to address complex problems in applied fields.

View unit details online (current students only)

View unit timetable

MXB161 Computational Explorations

Pre-requisites	Admission to (MS01 or MV01 or MV05 or MV06 or DS01 or IX30 or SE20 or SE30 or SE40 or SE70 or ST20 or SE07 or ID29) or 48 credit points of study in current course
Credit Points	12

This unit introduces you to techniques of computation and simulation across a range of application areas in Science, Technology, Engineering and Mathematics (STEM). Computation and simulation are cornerstones of modern practice across STEM; practitioners skilled in these areas can explore behaviours of real-world systems that would be impractical or impossible to undertake using only theoretical or experimental means. In this introductory unit, you will develop your computation and simulation skills through individual and collaborative problem-solving activities. Depending on your course, further exploration may be available through a minor in this field.

View unit details online (current students only)
View unit timetable

MXB201 Advanced Linear Algebra

Pre-requisites	MXB106 and (MXB102 or Admission to ST20)
Equivalents	MAB312
Credit Points	12

Much of the power of linear algebra stems from its widely-applicable collection of analytical tools for applied problem-solving. This unit builds upon your knowledge of linear algebra to explore more advanced techniques and applications of matrices and vectors. Furthermore, you will learn how much of what is familiar about linear algebra in Euclidean space can be abstracted to develop a more generally applicable theory. Hence you will develop an appreciation for the power and versatility of linear algebra across the mathematical sciences.

View unit details online (current students only)
View unit timetable

MXB202 Advanced Calculus

Pre-requisites	MXB105 and MXB106
Equivalents	MAB311
Credit Points	12

Advanced calculus is fundamental to the study of applied mathematics and related quantitative disciplines such as physics, physical chemistry and engineering. This unit introduces you to new skills and methodologies in multivariable and vector calculus that are essential to the study of science, technology and engineering, and it also provides you with the necessary background to go on to more advanced study in applied mathematics, such as partial differential equations and advanced mathematical modelling. This unit builds on your introductory calculus and linear algebra skills developed in MXB105 Calculus and Differential Equations and MXB106 Linear Algebra, and will further develop your ability to decompose complex problems into smaller components, resolve these smaller components and hence solve the original problem.

View unit details online (current students only)
View unit timetable

MXB225 Modelling with Differential Equations 1

Pre-requisites	MXB105
Equivalents	MAB413, MXB221
Credit Points	12

Differential equations are commonly used to formulate mathematical models of real-world phenomena from across science, engineering, economics and beyond. This unit builds on your earlier studies of differential equations to consider how such models are constructed, how to obtain analytical solutions, and how to use these models and their solution to gain insight into real-world processes.

View unit details online (current students only)
View unit timetable

MXB226 Computational Methods 1

Pre-requisites	MXB103 and MXB201
Equivalents	MAB420, MXB222
Credit Points	12

This is a foundational unit for Computational Mathematics. It introduces the design and implementation of computational techniques for solving a range of problems in mathematics. These techniques will be analysed for important properties such as efficiency, stability, convergence and error. The main topics that will be covered include: finite difference methods for models of heat diffusion in two dimensions; direct and iterative methods for linear systems; efficient storage of data; norms; approximation; numerical integration; numerical methods for ordinary differential equations.

View unit details online (current students only)
View unit timetable

MXB241 Probability and Stochastic Modelling 2

Pre-requisites	MXB101 and MXB105
Equivalents	MAB314
Credit Points	12

It is important to develop skills and knowledge in both statistics and mathematics. Building on the methodology and skills developed in previous studies in probability and stochastic modelling, this unit provides you with formal statistical tools such as stochastic process models and statistical methods for theoretical and applied development. These methods are useful in a wide range of areas, from communication systems and networks to traffic to law to biology to financial analysis, and link with other modern areas of mathematics. This unit will provide opportunities to learn how to build statistical models of real world processes, acknowledging the assumptions inherent in selected models. The skills developed in this unit will be integral in the understanding of material throughout your studies in statistics and mathematical modelling.

View unit details online (current students only)
View unit timetable

MXB242 Regression and Design

Pre-requisites	MXB107
Equivalents	MAB414
Credit Points	12

This is an intermediate applied statistics unit addressing the collection (design of experiments), exploration, summarisation, analysis and reporting of continuous data. You will analyse data using general linear models and communicate findings using oral and written methods. You will use mathematical and statistical software, such as R, to enhance your data analysis and develop your statistical programming skills. The application of statistical data analysis is pervasive across Engineering, Science, Health and Business. Hence, this unit is suitable for both Mathematics students and students in other disciplines. This unit is intended for students who have completed foundation studies in statistical data analysis and who wish to develop further skills in applied statistics. MXB344 Generalised Linear Models builds on this unit by considering the analysis of binary, categorical and count data. MXB343 Modelling Dependent Data extends this unit for data that are not independent.

View unit details online (current students only)
View unit timetable

MXB261 Game Theory and Simulation for Decision-Making

Equivalents	INB360, MAB480
Credit Points	12

With the growing importance of strategic decision-making in business and data science, understanding game theory is essential for analyzing competitive and cooperative interactions. This unit introduces the mathematical foundations of game theory and its applications in these fields. You will explore concepts such as Nash equilibrium, cooperative games, Bayesian games, and mechanism design, with a focus on computational techniques. Through hands-on simulations, you will model and analyze strategic interactions in real-world scenarios, including pricing strategies, auctions, network games, and resource allocation. The unit emphasizes the use of computational tools to study equilibrium behavior, optimization, and decision-making under uncertainty. By combining mathematical theory with computational practice, this unit equips you with the skills to design and analyze strategic decision-making models in business and data science applications.

View unit details online (current students only)
View unit timetable

MXB322 Partial Differential Equations

Pre-requisites	(MXB201 and MXB202) or MXB225 or MXB221 or MAB413
Equivalents	MAB613
Credit Points	12

Partial differential equations are the foundation of mathematical models that describe evolving processes exhibiting spatial and temporal variation. In this unit you will learn how the study of such equations synthesises and extends many of the concepts you have learned previously in linear algebra and calculus. The powerful frameworks of Fourier analysis and integral transforms that underpin partial differential equations provide a means for obtaining solutions to a number of equations of unparalleled physical importance, and for understanding the behaviour of mathematical models more generally.

View unit details online (current students only)
View unit timetable

MXB325 Modelling with Differential Equations 2

Pre-requisites	MXB322 and (MXB225 or MXB221)
Equivalents	MAB521, MXB321
Credit Points	12

Among the variety of differential equations encountered in applied mathematics, equations modelling the transport of quantities such as mass and energy are especially important. This unit significantly extends your repertoire by considering models with greater mathematical complexity than you have previously encountered, drawn from and representative of a variety of important real-world applications. Such complexity necessitates greater ingenuity in the analysis and solution of the governing equations, which will harness and extend your full knowledge of modelling with differential equations.

View unit details online (current students only)
View unit timetable

MXB326 Computational Methods 2

Pre-requisites	MXB202 and (MXB226 or MXB222)
Anti-requisites	EGB324
Credit Points	12

Advanced computational methods underpin essentially all modern computer simulations of complex real-world processes. This unit will significantly extend your toolset of computational methods, particularly for the solution of complex partial differential equation models of real phenomena. You will gain critical expertise and experience at building practical, efficient computer codes which will leverage advanced theoretical and algorithmic considerations that draw upon your full range of mathematical and computational knowledge and skills in linear algebra and calculus.

View unit details online (current students only)
View unit timetable

MXB332 Optimisation Modelling

Pre-requisites	MXB232 or MAB315
Equivalents	MAB525
Credit Points	12

Operations research techniques are used in numerous industries and are critical for decision making. These industries need graduates who can apply techniques of mathematical modelling, statistical analysis, mathematical optimisation and simulation and can implement these techniques using appropriate computer software packages. This unit will build upon the content of MXB232 by introducing more advanced “intermediate” level operations research methods and techniques. The topics addressed in this subject are vital in this field and are critical for advanced applications and studies in this field. Topics covered include: model building in mathematical programming, modelling language - (e.g. OPL, Gurobi or equivalent), integer programming and branch-and-bound method, introduction to inventory theory, dynamic programming; and computer solutions of advanced linear programming problems and their analysis.

View unit details online (current students only)
View unit timetable

MXB334 Operations Research for Stochastic Processes

Pre-requisites	MXB232 and MXB241
Equivalents	MAB625
Credit Points	12

This unit provides you with the opportunity to apply your knowledge and skills in operations research to guide decision-making for complex real-world problems. Your previous learning in deriving and solving operations research problems was mostly dealing with a decision making in a deterministic setting. The focus here is to optimize decision making when there is uncertainty and stochastic variables. Combined with the operations research expertise you have acquired over your degree, you will be able to formulate and solve these complex decision problems using computational tools.

View unit details online (current students only)
View unit timetable

MXB341 Statistical Inference

Pre-requisites	MXB241 or MAB314
Equivalents	MAB524
Credit Points	12

This is an advanced unit in mathematical statistics covering the theory of point estimation and inference using both classical and Bayesian methods. Statistical inference is the practice of both estimating probability distribution parameters and using statistical testing to validate these results, and plays a crucial role in research, and many real-world applications. You will use the methods of least squares, moments, and maximum likelihood to construct estimators of probability distribution parameters and evaluate them according to criteria including completeness, sufficiency, and efficiency. Results will be computed both analytically and numerically using software such as R. You will learn and apply the Neyman-Pearson Lemma for the construction of statistical tests, including to real-world applications, and learn Bayesian statistics for finding posterior distributions of parameters and evaluating their performance. Results will be communicated both orally and in written form.

View unit details online (current students only)
View unit timetable

MXB344 Generalised Linear Models

Pre-requisites	MXB242
Equivalents	MAB624
Credit Points	12

For data that arise in, for example, science and commerce, it is often unreasonable to assume they are continuous random variables from a normal distribution. It is likewise unlikely that data are handed to an analyst in a state ready for advanced statistical techniques. In this unit you will be introduced to modelling techniques and methodology for the explanation of non-normal data. You will also learn, by way of a realistic project, techniques to overcome common issues with shaping data for analysis. Hence, you will be well prepared in the application of appropriate statistical practice when such data are encountered in the real world.

View unit details online (current students only)
View unit timetable

MZB101 Modelling with Introductory Calculus

Equivalents	MAB105
Credit Points	12

This is an introductory unit covering the basic mathematical theory of functions and graphs, along with the foundational concepts and techniques of differential and integral calculus. The unit also explores a wide variety of calculus applications, and introduces the basic mathematical modeling skills relevant to a wide variety of scientific fields. The ability to express scientific problems in mathematical language, and to apply calculus techniques in the analysis of these problems, is essential to science students across all discplines. This introductory unit is particularly intended for students whose mathematical preparation does not include Queensland Senior Mathematics B or an equivalent. The mathematical foundation covered here will be developed further in SEB113.

View unit details online (current students only)
View unit timetable

PVB210 Stellar Astrophysics

Equivalents	PQB460
Credit Points	12

Astrophysics is the application of physics and chemistry to investigate the origin and evolution of galaxies, stars, planets, nebulae and other astronomical objects in the universe. Topics presented in this unit include Indigenous astronomy, orbital mechanics, telescopes, our solar system, the planets, star formation, stellar evolution, stellar remnants, and exoplanets.

View unit details online (current students only)
View unit timetable

PVB220 Cosmology

Equivalents	PQB660
Credit Points	12

Cosmology is the study of the origin and evolution of the universe, from the Big Bang to the present. Topics presented include special and general relativity, the physics and geometry of space-time, inflationary cosmology, cosmic microwave background, dark energy and dark matter, supermassive black holes, gravitational waves, and the status of Australian cosmology surveys.

View unit details online (current students only)
View unit timetable

QUT001 QUT You: Artificial Intelligence in the Real World

Credit Points	6

You engage with Artificial Intelligence (AI) daily, perhaps unknowingly, through search engines, social media, and streaming services. It is increasingly likely that you will apply AI tools in your future career, regardless of your field. But without any understanding of how AI systems are built and tested, how do you know that you can trust them? In this unit you will explore what AI is, how it's developed, and its potential pitfalls. You will learn to decipher AI jargon, to probe AI systems to understand their capabilities and limitations, consider how AI may impact your future career, and the challenges present in ensuring that AI acts in a fair and ethical manner. You will explore AI concepts and methods through interactive demonstrations and real-world tools which will demonstrate the possibilities and problems with AI, and its broad applicability across fields including Engineering, Business, Health, Law and Design.

View unit details online (current students only)
View unit timetable

QUT002 QUT You: Walking on Country

Credit Points	6

In this unit, you will explore the place in which QUT is situated, reflect on its culture both past and present, and discover knowledge embedded in place. Meeanjin, also known as Brisbane, is Country to the Turrbal and Yugara peoples, and home for many Aboriginal and Torres Strait Islander peoples, affectionately regarded as the Brisbane Blacks. This unit will allow you to ‘choose your own adventure’ to explore the varying ways in which to think of the significance of place, geologically, historically, culturally, and politically. You will experience an unsettling of the primacy of ‘Western’ framings of knowledge, environment, community and climate. In so doing, you will gain a deeper understanding of the oldest continuous living culture in the world, the sophistication and strength of Indigenous world views, and arrive at a fuller appreciation of the value of co-existing knowledge systems which can help us as we journey to a sustainable and racially just society.

View unit details online (current students only)
View unit timetable

QUT003 QUT You: Real Action for Real Change

Credit Points	6

Real world challenges such as climate change, health epidemics, data or food security are by their very nature complex. Solutions require both large scale changes through government and commercial sectors, but also locally driven action that is tailored to the nuances of a community. In this QUT You unit, you will develop an action plan related to a social, health, political or environmental real world challenge of interest. You will employ critical and creative thinking to drive positive impact. Generative AI technologies will be explored to further develop your change ideas. Your discoveries will identify the challenges, reasons for a lack of action, and likely areas of intervention that may result in a meaningful change to attitudes, technologies, and/or behaviour. Through supported classes, you will create a locally driven action plan to engage, educate, and encourage meaningful change from members of the general public or a relevant community.

View unit details online (current students only)
View unit timetable

QUT004 QUT You: Living and Working Collaboratively, Ethically, and Inclusively

Credit Points	6

We are living in an increasingly diverse, interconnected, and constantly evolving world. Understanding who we are and how our world view affects others is critical to our ability to work effectively with diverse teams and to make decisions that are good for everyone. Deliberative problem-solving and ethical decision-making are transferable skills that can be applied to our future studies, careers, and even our day-to-day lives. Through your active engagement in games, debates, and decision-making challenges drawn from a broad range of professional contexts, you will learn the knowledge and skills necessary to understand and resolve real world problems in ways that work for as many people as possible. The aim is to prepare you for life beyond university, both professionally and personally, so that you -- *our future leaders* -- can shape a better world for us all.

View unit details online (current students only)
View unit timetable

QUT005 QUT You: Seeing Me, Seeing You: Skills for a Diverse World

Credit Points	6

We both live and work in a diverse and multi-cultural society. To be successful in your professional and personal lives you need to recognise and appreciate the impact that your own identity and the identity of others contributes to our social interactions and relationships. This unit provides you with skills to become more culturally aware in your communication and express yourself in a confident, but respectful manner. By learning about the intricate nature of culture, language and power, and how they contribute to engagement with others, you will be better able to form meaningful and connected relationships. You will also learn to apply skills including observation, empathy, communication and social connection to understand and address power imbalances in personal and work relationships. These skills will enable you to work effectively in diverse teams and to foster inclusive environments in both personal and professional settings.

View unit details online (current students only)
View unit timetable

QUT006 QUT You: The Art of Pitching

Credit Points	6

Whether we are presenting a creative, experimental project, a scientific discovery, or business opportunity, pitching is a valuable, life-long skill that we typically learn through trial and error. Brilliant ideas don’t sell themselves. To survive and thrive in this globalised and competitive market, it’s vital to be able to communicate our ideas concisely, creatively, and persuasively to diverse audiences and gain their support. In this unit you will learn how to develop and deliver a pitch that inspires diverse audiences and leaves a lasting positive impression. Researching topics that have personal and real-world significance, you will be guided by the teaching team and learn how to concisely and confidently communicate your ideas in innovative and compelling ways. The knowledge and skills developed in this unit have application across academic and professional contexts.

View unit details online (current students only)
View unit timetable

QUT007 QUT You: Fighting 'Fake News'

Credit Points	6

The modern world is a minefield of misinformation. Without a suite of skills to critique the information we receive, we are all potentially prey to deception and misinformation from a variety of sources, including academic, media and social sources. In this unit, you will learn how to evaluate the reliability of information to make informed decisions. You will learn to identify hidden agendas, biases, and influences behind the messages we receive. You will question your own views, their origins, and their ongoing sources of influence. This unit will equip you to make informed decisions and take responsible actions. This involves critical self-reflection, and an understanding of the tricks that can be used to bypass your usual critical thinking skills. You will also develop a set of criteria on which the value of evidence and arguments can be judged. Media and technology use will be a particular focus, as we examine the personal influences which shape our views.

View unit details online (current students only)
View unit timetable

QUT008 QUT You: Think Like a Computer and Change the World

Credit Points	6

Whether you dream of becoming an environmental scientist, an interaction designer, a health professional, or a market analyst, your future career will require you to use computers to deal with harder problems and more data about more things than ever before. In this hands-on unit you will learn and apply computational thinking – the art of formulating problems in such a way that a machine can solve them for you. Each week you will tackle a range of real world and interdisciplinary challenges using simple coding resources and techniques. You will also work collaboratively to evaluate and critique the use and application of coding techniques in different disciplines to understand their societal implications. At the end of the unit, you will be able to apply these skills and knowledge to your own discipline to better prepare you for our increasingly technology-driven world. You will understand how coding works, and you will know how to critique different approaches.

View unit details online (current students only)
View unit timetable

QUT009 QUT You: Data Science for Society

Credit Points	6

Data is part of the fabric of our modern societies with almost all aspects of our lives influenced, for better or worse, by systems that are fundamentally data-driven. As individuals, we often unknowingly contribute enormous quantities of data to these systems through our use of smart devices, wearables, and online platforms. Understanding the power and limitations of the rapidly growing field of data science is more important than ever before. In this unit, you will identify sources of bias, error, and misinterpretation within the data science pipeline and the potential consequences of data-driven decision-making if these sources are left unchecked. This grounding in fundamental principles of data science will empower you to think critically and ethically about these systems and how they affect us. Regardless of your career or discipline, you have a role to play in ensuring data-driven systems are built that align with our personal values and the values of our society.

View unit details online (current students only)
View unit timetable

QUT010 QUT You: People with Robots

Credit Points	6

What does the future look like in a world where robots provide support and care to older people, drive public transport, work alongside us in health and manufacturing settings, or greet us at a restaurant? Distinct from industrial robots, collaborative robots are uniquely designed to be easy to use and safe for people to work with. The ways in which collaborative robots are designed, used, and adopted can have significant impact on future societies. In this unit we will focus on physical robots that interact with people. We will design and evaluate collaborative robot concepts in response to real world challenges and examine their impact on people and robot collaboration. We will imagine what an ethical, sustainable, and socially just future could look like where people and robots work together in relation to your future career. In doing so you will be better equipped to engage with and critique the role that collaborative robots might have in your life.

View unit details online (current students only)
View unit timetable

QUT013 QUT You: Human Agency in the Digital World

Credit Points	6

In a world where so much of our activity is in digital environments, it is vital to understand how digital systems influence our choices, attention and behaviour, and how we can stay in control of them. This unit helps you recognise the psychological processes that shape your digital experiences and how digital tools can both support and undermine your personal and professional goals. You'll explore social media, games, and online marketing to understand their risks, opportunities and real potential for positive use. Through handson and collaborative activities, you'll learn to navigate digital environments on your own terms and design habits that support your values, wellbeing and future professional practice.

View unit details online (current students only)
View unit timetable

SCB300 Professional Practice

Pre-requisites	Successfully completed of 96cp of coursework units and admission into IN01 or IN05 or ST01 or DS01 or MS01)
Other requisites	Placements must be pre-approved by the Faculty of Science WIL team.
Assumed Knowledge	null
Credit Points	12

This unit provides you with the opportunity to further develop your professional capabilities and technical skills by undertaking the equivalent of 15 - 30 day full time paid or unpaid work placement. This unit, is typically undertaken in the penultimate or final year of your course, requiring you to undertake a self-sourced work placement, internship or engage in ongoing professional work experience to consolidate your course-based learning. In this unit you will have the opportunity to refine your reflective and communicative skills, formulating effective statements of your experience suitable for inclusion in job applications. Assessments in this unit require you to critically analyse, report and reflect upon your professional practice experience and to prepare a professional development plan to help guide your career.

View unit details online (current students only)
View unit timetable